PHA Study Report – Ammonia unit – reforming, synthesis, refrigeration and purge recovery

Fertiliser complex (illustrative example)

Worked example for the Process Safety Analysis Toolkit: HAZID · What-If / Checklist · HAZOP · FMECA · Fault tree · Event tree · Bow tie

Item	Detail
Documents	HAZID HI-EX-AM-001 · What-If WI-EX-AM-001 · HAZOP HZ-EX-AM-001 · FMECA FM-EX-AM-001
Study dates	2026-11-12 (HAZID) · 2026-11-13 (What-If) · 2026-11-16 to 2026-11-20 (HAZOP) · 2026-11-23 (FMECA)
Facilitator / scribe	A. Facilitator / B. Scribe (Istec)
Revision	A – worked example

Illustrative training example. The plant arrangement, tag numbers and failure data are representative of a natural-gas-based ammonia unit at a fertiliser complex; they are not design data of any real plant and must not be used for decisions about one.

1 Purpose and how to use this example

This example carries one study through every tool of the Process Safety Analysis Toolkit for a large ammonia unit: the HAZID screens the unit and picks the HAZOP nodes; the HAZOP finds the scenarios and the safeguards claimed; the FMECA looks at how the protective equipment fails; the fault trees quantify the protection layers; two event trees follow a toxic and a flammable release to their outcomes; two bow ties show the barriers to manage. Every file opens in its tool, and the numbers agree across them.

Key results: refrigeration receiver over-pressured beyond design 2.4E-6 /yr (fault tree FT-AM-01, protection PFDavg 1.1E-5); converter over-temperature beyond the protection 1.8E-2 /yr (FT-AM-02, PFDavg 5.1E-2); ammonia detection and ESD PFDavg 2.9E-2 (FT-AM-03); toxic fatalities on and off site 1.3E-6 /yr (ETA-AM-01); vapour cloud explosion at the synthesis loop 8.1E-5 /yr (ETA-AM-02).

File (Examples / Ammonia Unit PHA)	Open with	Content
Ammonia unit - HAZID_Register.xlsx	Excel	HAZID register – 22 hazards
Ammonia unit - What-If WI-EX-AM-001.whatif.json	What-If Checklist Analysis	What-If / Checklist – 4 areas, 23 questions
Ammonia unit - WhatIf_Checklist_Workbook.xlsx	Excel	The same What-If study in the workbook
Ammonia unit - HAZOP HZ-EX-AM-001.hazop.json	HAZOP Study Manager	5 nodes, 42 deviations, 20 recommendations
Ammonia unit - HAZOP_Study_Workbook.xlsx / .xlsm	Excel	The same HAZOP in the workbook
Ammonia unit - FMECA_Workbook.xlsx / .xlsm	Excel	FMECA – 5 systems, 17 failure modes
Ammonia unit - FT-AM-01 Refrigeration overpressure.fta.json	Fault Tree & PFD Calculator	FT-AM-01 – Ammonia refrigeration receiver over-pressured beyond design
Ammonia unit - FT-AM-02 Converter over-temperature.fta.json	Fault Tree & PFD Calculator	FT-AM-02 – Uncontrolled over-temperature of the ammonia converter 105-D
Ammonia unit - FT-AM-03 Ammonia detection and ESD.fta.json	Fault Tree & PFD Calculator	FT-AM-03 – Ammonia release at the refrigeration area not isolated
Ammonia unit - ETA-AM-01 Liquid ammonia release.eta.json	Event Tree Analysis	ETA-AM-01 – Liquid ammonia release from the refrigeration system
Ammonia unit - ETA-AM-02 Synthesis gas release.eta.json	Event Tree Analysis	ETA-AM-02 – Synthesis gas release at the synthesis loop
Ammonia unit - BT-AM-01 Ammonia loss of containment.bowtie.json	Bow Tie Analysis	BT-AM-01 – Loss of containment of liquid ammonia from the refrigeration and recovery systems
Ammonia unit - BT-AM-02 Synthesis gas loss of containment.bowtie.json	Bow Tie Analysis	BT-AM-02 – Loss of containment of hydrogen-rich synthesis gas at the synthesis loop
Ammonia unit - LOPA LP-EX-AM-001.lopa.json	LOPA Study Manager	LOPA LP-EX-AM-001 – 11 scenarios from the HAZOP LOPA study list (section 5.2)

2 Source study and how this example was developed

The starting point is a published HAZOP report of a natural-gas-based ammonia unit. It covers about 25 nodes with deviation, cause, consequence, safeguard and recommendation columns and numbered recommendations, but no risk ranking and no quantification. The plant owner, licensor and contractor are replaced by a fictional fertiliser complex; the source tag numbers and set points are kept.

Five nodes are taken forward – source nodes 3 (reforming), 23 (reformer fuel gas), 15 (syngas compressor and converter), 17 (refrigeration) and 18 (purge-gas recovery). Source entries are marked 'source'; gaps are filled from good practice for ammonia plants. Risk ranking, the FMECA and the quantified studies are added.

2.1 Gaps filled and corrections

#	Gap or weakness in the source	What this example does	Where
1	The source worksheets have no severity or likelihood	Every deviation risk-ranked on the 5 x 5 matrix before and after recommendations	All nodes
2	Refrigeration overpressure relies on PSHH-1110 on the same transmitter as the controller	Modelled as a common-cause event; independent transmitter recommended	N04-01, FT-AM-01, R-011
3	Converter high-temperature alarm and trip on one thermocouple	Second independent thermocouple, 1oo2 trip; quantified	N03-03, FT-AM-02, R-008
4	Ammonia release: 'gas detectors provided' without coverage or response	Detector voting, coverage and ESD response quantified; water curtain and shelter as event-tree headings	N04-07, FT-AM-03, ETA-AM-01, R-012, R-013, R-016
5	No toxic-refuge or HVAC trip for the control room	Added as a barrier and event-tree heading	BT-AM-01 SHELTER, R-016
6	Hydrogen release treated like natural gas	Higher ignition probabilities for hydrogen; VCE in the congested compressor house	ETA-AM-02, BT-AM-02
7	High-temperature hydrogen attack not considered	Added as a deviation and threat with an inspection barrier	N03-06, R-010
8	Gas blow-by from the 140 barg loop into the 20 barg column not checked	Relief sizing for blow-by recommended	N05-02, R-018

3 Scope and system description

Five of the 27 nodes of the ammonia unit: N01 (source node 3) desulfurised feed gas with steam through the mixed-feed preheat coil to the primary reformer 101-B and the secondary reformer 103-D; N02 (source node 23) fuel gas to the primary-reformer arch burners 101-BBA, tunnel burners 101-BBT and superheater burners 101-BBS; N03 (source node 15) synthesis gas compressor 103-J and ammonia converter 105-D; N04 (source node 17) ammonia refrigeration – compressor 105-J, unitized chiller 120-C with flash drums 120-CF1…CF4, condenser 127-C and receiver 149-D; N05 (source node 18) loop purge-gas ammonia recovery with the ammonia distillation column 160-C.

Design basis: Reform natural gas with steam (steam-to-carbon ratio 3.0) at about 33 barg and 800 °C, make synthesis gas, compress it to about 140 barg and convert it to ammonia in 105-D; condense and refrigerate the product with an ammonia refrigeration system (flash drums 120-CF1/CF2 designed for 15.5 barg and 0.25 bar vacuum – source 15.8 kg/cm2G); recover ammonia from the loop purge. All values are illustrative.

Exclusions: The other 22 nodes of the source study (front-end, CO shift, CO2 removal, methanator, steam, drains); ammonia storage tanks; urea plant. Licensor proprietary design details.

Node	Name	Design intent	Design / operating parameters
N01	Source node 3 – Feed gas and steam to the primary reformer 101-B and secondary reformer 103-D	Reform desulfurised natural gas with steam over nickel catalyst at a steam-to-carbon ratio of 3.0, and complete the reforming with process air in 103-D	101-B tubes: design 800 °C metal / 41 barg; TAH-1325 tube outlet; FFSLL-1201 trips at steam-to-carbon 2.5; PSHH-1059 high pressure Operating: Inlet 38 barg, outlet 33 barg; reformer outlet 800 °C; 103-D outlet 1 000 °C
N02	Source node 23 – Fuel gas to the primary reformer arch burners 101-BBA, tunnel burners 101-BBT and superheater burners 101-BBS	Deliver fuel gas to the reformer burners only when the furnace is purged, combustion air is proven and flames are proven, and shut it off on any unsafe condition	Fuel gas header 6 barg design; PSLL-1221 0.3 barg; PSHH-1221 2.5 barg; firebox pressure trip PAHH-1226 (loss of draught) Operating: Burner pressure 0.5–2.0 barg; firebox −5 mmWC; about 400 burners
N03	Source node 15 – Synthesis gas compressor 103-J and ammonia converter 105-D	Compress synthesis gas to about 140 barg and convert hydrogen and nitrogen to ammonia over iron catalyst at 380–520 °C without exceeding the loop design	Loop design 160 barg / 400 °C (converter shell); converter catalyst 520 °C max; TSHH-1396 trips 103-J and closes the converter feed; PSV-1018 on the loop at 155 barg Operating: Loop 135–145 barg; converter outlet 450 °C; 103-J speed 10 000 rpm
N04	Source node 17 – Ammonia refrigeration: 105-J, unitized chiller 120-C, flash drums 120-CF1…CF4, condenser 127-C, receiver 149-D	Refrigerate the synthesis loop and condense product ammonia, keeping the flash drums within their pressure and level limits and liquid out of 105-J	Flash drums 120-CF1/CF2 design 15.5 barg and 0.25 bar vacuum (source 15.8 kg/cm2G); receiver 149-D 20 barg, PSV-1109 at 20 barg; LSHH-1214… 1217 trip 105-J; PSHH-1110 high discharge pressure trip; design minimum temperature −33 °C Operating: Flash drums −33 to +10 °C; condenser 127-C 16–18 barg; about 4 000 t refrigerant inventory hold-up 120 t
N05	Source node 18 – Loop purge-gas ammonia recovery and ammonia distillation column 160-C	Recover ammonia from the loop purge so that the purge gas sent to fuel is ammonia-free, and return the ammonia to the product	Scrubber 124-D 160 barg; column 160-C 20 barg, PSV-1033 at 20 barg; FAL-1064 low wash-water flow Operating: Purge 140 barg reduced to 20 barg; column 18 barg

3.1 Risk criteria

All studies use the toolkit's 5 x 5 risk matrix: risk = severity (1–5) x likelihood (1–5); Low < 5, Medium 5–9, High 10 –19, Very High ≥ 20. Likelihood bands: 1 < 1E-4 /yr, 2 1E-4–1E-3, 3 1E-3–1E-2, 4 1E-2–0.1, 5 > 0.1 /yr. Event-tree and bow-tie targets: catastrophic outcomes ≤ 1E-4 /yr, major ≤ 1E-2 /yr.

3.2 Team and sessions

Name	Company / department	Role	Attendance
A. Facilitator	Istec	Facilitator	Days 1–5
B. Scribe	Istec	Scribe	Days 1–5
C. Process	Owner	Process engineer (ammonia)	Days 1–5
D. Operations	Owner	Ammonia plant shift supervisor	Days 1–5
E. Instrument	EPC contractor	Instrument / SIS engineer	Days 1–5
F. Mechanical	EPC contractor	Rotating equipment engineer	Days 3–4
G. Safety	Owner	HSE / process safety	Days 1–5
H. Process	EPC contractor	Process engineer	Days 1–5
I. Vendor	Compressor vendor	Compressor specialist	Day 3

Date	Nodes	Notes
2026-11-16	N01	Node 3 (primary and secondary reforming) – source worksheet reviewed and risk-ranked; tube rupture and steam-to-carbon trips.
2026-11-17	N02	Node 23 (reformer fuel gas and BMS) – firebox explosion and light-off.
2026-11-18	N03	Node 15 (syngas compressor and converter) – converter over-temperature referred to FT-AM-02 and BT-AM-02; releases to ETA-AM-02.
2026-11-19	N04	Node 17 (refrigeration) – receiver overpressure referred to FT-AM-01; ammonia release to FT-AM-03, ETA-AM-01 and BT-AM-01.
2026-11-20	N05	Node 18 (purge-gas recovery) – gas blow-by into 160-C. Study closed; FMECA on 23–24 Nov.

4 HAZID and What-If / Checklist

4.1 HAZID register

Structured brainstorm with the HAZID guidewords; S x L on the toolkit's 5 x 5 risk matrix; High / Very High items sent to a HAZOP node or a quantified study. The whole ammonia unit – reforming, shift, CO2 removal, synthesis, refrigeration, purge recovery, utilities – to select the nodes for the detailed HAZOP.

ID	Area	Guideword	Hazard	Cause / threat	Consequence	Existing safeguards	S x L	Further study	Rec.
HZD-01	Reforming	Fire and explosion	Reformer tube rupture	Low steam-to-carbon, over-firing, creep	Syngas fire in the 101-B firebox	FFSLL-1201, FSLL-1201 trips; tube surveys	4 x 3 = 12 High	HAZOP N01-01, N01-04	R-001, R-002
HZD-02	Reforming	Temperature (high)	Secondary reformer over-temperature	Excess process air	Refractory and shell damage, hot spots	FFSHH-1203 trip	4 x 3 = 12 High	HAZOP N01-06	R-003
HZD-03	Reforming	Fire and explosion	Firebox explosion	Flame-out with fuel flowing; light-off without purge	Furnace casing failure, fatalities	BMS, purge interlock	5 x 3 = 15 High	HAZOP N02-01, N02-06	R-004
HZD-04	Reforming	Fire and explosion	Fuel-gas leak at burner manifolds	Flexible hose / flange failure	Burner-platform fire	PTW, PPE	4 x 3 = 12 High	HAZOP N02-08	
HZD-05	Reforming	Process upset	Ammonia in fuel gas	Carry-over from purge recovery	NOx, flame instability	Knock-out drum, scrubber	3 x 3 = 9 Medium	HAZOP N02-04	R-005
HZD-06	Synthesis	Fire and explosion	Syngas release at the compressor or loop	Seal failure, flange leak, HTHA	Hydrogen jet fire, VCE	ESD, depressurisation	5 x 3 = 15 High	HAZOP N03-08; ETA-AM-02; BT-AM-02	R-019
HZD-07	Synthesis	Temperature (high)	Converter over-temperature	Quench control failure, ratio upset, start-up heater	Converter / exchanger failure	TAH-1383, TSHH-1396	5 x 4 = 20 Very High	HAZOP N03-03; FT-AM-02; BT-AM-02 T1	R-008
HZD-08	Synthesis	Pressure	Loop overpressure	Blocked outlet, high inerts	Flange failure, large release	PSV-1018	5 x 3 = 15 High	HAZOP N03-04	R-009
HZD-09	Synthesis	Materials	High-temperature hydrogen attack	Operation outside the Nelson-curve limits	Sudden exchanger failure	Material selection	5 x 2 = 10 High	HAZOP N03-06	R-010
HZD-10	Synthesis	Mechanical	Compressor surge / damage	Strainer choking, anti-surge failure	Seal damage, release	Anti-surge control	4 x 3 = 12 High	HAZOP N03-02	R-006, R-007
HZD-11	Refrigeration	Toxicity	Large liquid ammonia release	Overpressure, seal / flange failure, maintenance error	Toxic cloud on and off site	Detection, ESD, water curtain	5 x 3 = 15 High	HAZOP N04-01, N04-07; FT-AM-01; FT-AM-03; ETA-AM-01; BT-AM-01	R-011, R-012, R-013
HZD-12	Refrigeration	Pressure	Receiver overpressure	Loss of cooling water to the condenser	Relief to flare; rupture if relief fails	PAH-1009, PSHH-1110, PSV-1109	5 x 4 = 20 Very High	HAZOP N04-01; FT-AM-01	R-011, R-015
HZD-13	Refrigeration	Temperature (low)	Brittle fracture	Auto-refrigeration on depressurisation	Sudden ammonia release	Low-temperature steel	5 x 2 = 10 High	HAZOP N04-05	R-014
HZD-14	Refrigeration	Mechanical	Liquid carry-over to the refrigeration compressor	High flash-drum level	Compressor damage, seal leak	LSHH-1214…1217	4 x 3 = 12 High	HAZOP N04-02	
HZD-15	Refrigeration	Maintenance	Line break with ammonia inside	Incomplete draining	Chemical burns, inhalation	PTW, PPE	4 x 3 = 12 High	HAZOP N04-08	
HZD-16	Purge recovery	Pressure	Gas blow-by into the distillation column	Let-down valve fails open	Column rupture – toxic release	PSV-1033	5 x 3 = 15 High	HAZOP N05-02, N05-04; BT-AM-01 T6	R-018
HZD-17	Purge recovery	Toxicity	Ammonia leak at the recovery pumps	Seal failure	Toxic exposure	Detection, PPE	4 x 3 = 12 High	HAZOP N05-06	R-012
HZD-18	Whole unit	Emergency response	Toxic-gas emergency	Large ammonia release	Exposure of control-room and plant staff	Refuge, escape sets	5 x 2 = 10 High	ETA-AM-01 H4	R-016
HZD-19	Whole unit	Utilities	Loss of instrument air / power	Compressor trip, grid failure	Plant trip; restart hazards	Fail-safe valves, UPS	3 x 3 = 9 Medium	HAZOP N01-08, N05-07	
HZD-20	Whole unit	Natural events	Lightning / storms	Summer storms	Ignition at vents; power loss	Lightning protection	3 x 2 = 6 Medium	–	
HZD-21	Whole unit	Environment	Ammonia and NOx emissions	Relief to flare, ammonia in fuel	Permit exceedance	Flare, scrubber	3 x 3 = 9 Medium	HAZOP N04-09, N02-04	R-005
HZD-22	Whole unit	Security	Unauthorised access	Contractor traffic	Tampering with valves	Access control	3 x 2 = 6 Medium	–	

4.2 What-If / Checklist analysis

Method: combined What-If / Checklist analysis (CCPS, Guidelines for Hazard Evaluation Procedures, 3rd ed., section 5.2), study WI-EX-AM-001, 2026-11-13. It follows the HAZID HI-EX-AM-001 and prepares the HAZOP HZ-EX-AM-001: the team listed What-If questions for each area or task, used the checklist to cover any gaps, and then decided for each question whether it is closed at What-If level, sent to a HAZOP node, or sent to a quantified study.

Three non-routine tasks the HAZID flagged – refrigeration maintenance, synthesis-loop start-up and reformer light-off – and the toxic-gas readiness of the refrigeration and purge-recovery areas. Questions that need a systematic deviation analysis are sent to HAZOP nodes N01–N05.

No.	Area / function / task	Type	Checklist packs	Checklist answered	Gaps (No / ?)	What-Ifs	Feeds HAZOP
A01	Refrigeration system maintenance – draining, line breaks, PSV removal	Task / procedure	GEN, NH3	8	3	6	N04
A02	Synthesis loop start-up	Task / procedure	GEN, NH3	5	2	6	N03
A03	Reformer light-off and steam-feed sequence	Task / procedure	GEN, NH3	4	1	4	N01, N02
A04	Toxic-gas readiness – refrigeration and purge-recovery areas, control room	Physical area	GEN, NH3	7	4	7	N04, N05

Checklist gaps (answered No or ?) and the What-If question each raised:

Area	Item	Checklist question	Answer	Note	What-If
A01	NH3-OPS-01	Is ammonia draining done by a closed method to a safe destination?	No	Open drain used in the past	A01-01
A01	NH3-MNT-01	Is equipment that held ammonia drained, purged and checked before a line break?	?	MP-AM-11 silent on trapped liquid	A01-02
A01	NH3-MNT-02	Are relief valves on ammonia service removed and refitted under a controlled procedure?	No	PSV removed with receiver in service	A01-03
A02	NH3-INS-01	Are the converter temperature alarm and trip on independent thermocouples?	No	Alarm and trip on one thermocouple	A02-02
A02	NH3-MAT-03	Is high-temperature hydrogen attack considered for synthesis-loop equipment?	?	Start-up temperatures not checked against the Nelson curve	A02-03
A03	NH3-SSD-02	Is the reformer light-off controlled by a purge interlock and a light-off procedure?	?	Purge timed by the operator	A03-01
A04	NH3-EMR-01	Is there a water curtain or spray system to knock down ammonia vapour, and is it tested?	?	Manual start only	A04-03
A04	NH3-EMR-02	Does the control room have a toxic-gas refuge with HVAC trip on ammonia?	No	No HVAC trip	A04-04
A04	NH3-EMR-03	Are escape sets and gas-tight suits available and people trained to use them?	?	Escape sets in one cabinet only	A04-05
A04	NH3-INS-02	Are ammonia detectors placed to cover the receivers, pumps and compressor house?	No	No detector at the pump bay	A04-02

What-If worksheet – A01 Refrigeration system maintenance – draining, line breaks, PSV removal (task / procedure)

Ref	What if…?	Consequence	Safeguards	Risk before	Recommendation	Risk after	Next step
A01-01	What if liquid ammonia is drained to the open sewer?	Toxic vapour in the area; ammonia to the effluent	• Permit-to-work	4 x 3 = 12	WR-01 Drain ammonia only through the closed drain system to the recovery drum; blank the open drain connections.	4 x 1 = 4	Closed at What-If
A01-02	What if a line is broken with liquid ammonia still trapped between closed valves?	Ammonia spray onto the maintenance crew	• Permit-to-work with positive isolation • Ammonia PPE	4 x 3 = 12	WR-03 Add a trapped-liquid check (bleed and wait) to MP-AM-11 before any flange is broken on ammonia service.	4 x 2 = 8	Send to HAZOP → N04-08
A01-03	What if PSV-1109 is removed for testing while the receiver 149-D is in service?	Overpressure of 149-D not relieved	• PSHH-1110 – High-high pressure trip [IPL]	5 x 2 = 10	WR-02 Remove PSV-1109 only with the receiver isolated and depressurised, or fit a spare PSV on a changeover valve.	5 x 1 = 5	Closed at What-If
A01-04	What if cooling water to 127-C is lost while 105-J is restarted after	Receiver pressure rises to the relief set point	• PAH-1009 – High-pressure alarm [IPL]	5 x 3 = 15		5 x 1 = 5	Send to quantified study (FT / ETA / bow

Ref	What if…?	Consequence	Safeguards	Risk before	Recommendation	Risk after	Next step
	maintenance?		• PSHH-1110 – High-high pressure trip [IPL] • PSV-1109 – Receiver relief valve [IPL]				tie) → FT-AM-01
A01-05	What if a drained section is depressurised quickly and the pipe gets very cold?	Brittle fracture of carbon steel	• Low-temperature steel [IPL]	5 x 2 = 10		5 x 1 = 5	Send to HAZOP → N04-05
A01-06	What if a flash drum is refilled too fast after maintenance and liquid reaches 105-J?	Compressor damage and seal leak	• LSHH-1214 – Flash-drum high-high level trip [IPL]	4 x 3 = 12		4 x 2 = 8	Send to HAZOP → N04-02

What-If worksheet – A02 Synthesis loop start-up (task / procedure)

Ref	What if…?	Consequence	Safeguards	Risk before	Recommendation	Risk after	Next step
A02-01	What if the start-up heater 102-B fires without loop circulation?	Local overheating of the converter and the loop	• Start-up procedure SU-AM-02	5 x 3 = 15		5 x 2 = 10	Send to HAZOP → N03-09
A02-02	What if the converter temperature alarm and trip read the same thermocouple?	Converter over-temperature not stopped	• TAH-1383 – Bed temperature alarm • TSHH-1396 – High-high temperature trip [IPL]	5 x 4 = 20	WR-04 Add a second, independent converter bed thermocouple for the high-high trip (see HAZOP R-008).	5 x 2 = 10	Send to quantified study (FT / ETA / bow tie) → FT-AM-02
A02-03	What if the loop exchanger runs above its design temperature during start-up?	Sudden exchanger failure – syngas release	• Material selection to API RP 941	5 x 2 = 10	WR-05 Hold the loop exchanger below its Nelson-curve limit during start-up; add a temperature alarm on the exchanger shell.	5 x 1 = 5	Send to HAZOP → N03-06
A02-04	What if 103-J surges during the run-up?	Seal damage – syngas release	• Anti-surge control • PSLL-1155 – Seal-gas low-low pressure trip [IPL]	4 x 3 = 12		4 x 2 = 8	Send to HAZOP → N03-02
A02-05	What if the loop is pressured up with the purge to recovery closed?	Loop overpressure – flange leak	• PSV-1018 – Synthesis loop relief valve [IPL]	5 x 2 = 10		5 x 1 = 5	Send to HAZOP → N03-04
A02-06	What if a loop flange leaks during heat-up?	Jet fire or explosion in the compressor house	• HV-1019 – Emergency depressurisation • UCP-1000 – Unit ESD [IPL]	5 x 3 = 15		5 x 2 = 10	Send to quantified study (FT / ETA / bow tie) → ETA-AM-02

What-If worksheet – A03 Reformer light-off and steam-feed sequence (task / procedure)

Ref	What if…?	Consequence	Safeguards	Risk before	Recommendation	Risk after	Next step
A03-01	What if a burner is lit before the firebox purge is complete?	Firebox explosion	• BSLL-1200 – Flame scanners and BMS [IPL] • Light-off procedure SU-AM-01	5 x 3 = 15	WR-06 Make the firebox purge a hard permissive in the BMS; light-off only with the purge-complete lamp on.	5 x 1 = 5	Send to HAZOP → N02-06
A03-02	What if feed gas is admitted before steam is established through the tubes?	Carbon on the catalyst – tube overheating and rupture	• FFSLL-1201 – Low steam-to-carbon trip [IPL] • XV-1201 – Feed shut-off valve	4 x 3 = 12		4 x 2 = 8	Send to HAZOP → N01-09
A03-03	What if process air is introduced too early or too fast?	Secondary reformer over-temperature	• FFSHH-1203 – High air-to-feed trip [IPL]	4 x 3 = 12		4 x 2 = 8	Send to HAZOP → N01-06
A03-04	What if a burner hose leaks while the burners are lit by hand?	Flash fire at the burner platform	• Hose inspection before light-off • Fire-resistant clothing	4 x 3 = 12		4 x 2 = 8	Send to HAZOP → N02-08

What-If worksheet – A04 Toxic-gas readiness – refrigeration and purge-recovery areas, control room (physical area)

Ref	What if…?	Consequence	Safeguards	Risk before	Recommendation	Risk after	Next step
A04-01	What if a large ammonia leak occurs at the receiver 149-D?	Toxic cloud on and off site	• AT-1801A/B/C – Ammonia detection and ESD [IPL] • Water-spray curtain	5 x 3 = 15		5 x 2 = 10	Send to quantified study (FT / ETA / bow tie) → ETA-AM-01

Ref	What if…?	Consequence	Safeguards	Risk before	Recommendation	Risk after	Next step
A04-02	What if the ammonia detectors do not cover the pump bay?	ESD not started; larger toxic cloud	• Operator rounds	5 x 3 = 15	WR-10 Carry out ammonia detector mapping for the pump bay and the compressor house.	5 x 2 = 10	Send to quantified study (FT / ETA / bow tie) → FT-AM-03
A04-03	What if the water curtain does not start during an ammonia release?	Vapour not knocked down; cloud reaches occupied areas	• Manual start from the control room	5 x 3 = 15	WR-08 Start the water curtain automatically on two ammonia detectors (2oo3).	5 x 2 = 10	Send to quantified study (FT / ETA / bow tie) → ETA-AM-01
A04-04	What if ammonia enters the control room through the ventilation?	Control-room staff overcome; plant left unattended	• Escape sets in the control room	5 x 2 = 10	WR-07 Trip the control-room HVAC on ammonia detected at the air intake.	5 x 1 = 5	Closed at What-If
A04-05	What if people cannot reach escape sets during a release?	Exposure above IDLH while escaping	• Ammonia drills	5 x 2 = 10	WR-09 Place escape sets at every exit of the compressor house and the control room; check them weekly.	5 x 1 = 5	Closed at What-If
A04-06	What if the purge let-down valve PV-1033A fails open?	Column over-pressured – toxic release	• PSV-1033 – Column relief valve [IPL]	5 x 3 = 15		5 x 2 = 10	Send to HAZOP → N05-02
A04-07	What if a seal leaks on the reflux pump 161-J?	Toxic exposure of the operator	• Gas detection • Ammonia PPE	4 x 3 = 12		4 x 2 = 8	Send to HAZOP → N05-06

Outcome: 4 questions closed at What-If level, 13 sent to the HAZOP, 6 sent to quantified studies, 0 need further information. 10 What-If recommendations (WR-xx) are tracked with the HAZOP recommendations.

HAZID	Hazard	S x L	Examined in What-If	Sent on to
HZD-01	Reformer tube rupture	4 x 3	A03-02	N01-09
HZD-02	Secondary reformer over-temperature	4 x 3	A03-03	N01-06
HZD-03	Firebox explosion	5 x 3	A03-01	N02-06
HZD-04	Fuel-gas leak at burner manifolds	4 x 3	A03-04	N02-08
HZD-06	Syngas release at the compressor or loop	5 x 3	A02-06	ETA-AM-02
HZD-07	Converter over-temperature	5 x 4	A02-01, A02-02	N03-09, FT-AM-02
HZD-08	Loop overpressure	5 x 3	A02-05	N03-04
HZD-09	High-temperature hydrogen attack	5 x 2	A02-03	N03-06
HZD-10	Compressor surge / damage	4 x 3	A02-04	N03-02
HZD-11	Large liquid ammonia release	5 x 3	A04-01, A04-02, A04-03	ETA-AM-01, FT-AM-03
HZD-12	Receiver overpressure	5 x 4	A01-03, A01-04	FT-AM-01
HZD-13	Brittle fracture	5 x 2	A01-05	N04-05
HZD-14	Liquid carry-over to the refrigeration compressor	4 x 3	A01-06	N04-02
HZD-15	Line break with ammonia inside	4 x 3	A01-01, A01-02	N04-08
HZD-16	Gas blow-by into the distillation column	5 x 3	A04-06	N05-02
HZD-17	Ammonia leak at the recovery pumps	4 x 3	A04-02, A04-07	FT-AM-03, N05-06
HZD-18	Toxic-gas emergency	5 x 2	A04-04, A04-05	

5 HAZOP results

Node	Name	Deviations	Highest risk before	Highest risk after	Recommendations
N01	Source node 3 – Feed gas and steam to the primary reformer 101-B and secondary reformer 103-D	9	12	8	R-001, R-002, R-003
N02	Source node 23 – Fuel gas to the primary reformer arch burners 101-BBA, tunnel burners 101-BBT and superheater burners 101-BBS	8	15	10	R-004, R-005
N03	Source node 15 – Synthesis gas compressor 103-J and ammonia converter 105-D	9	20	20	R-006, R-007, R-008, R-009, R-010, R-019, R-020

Node	Name	Deviations	Highest risk before	Highest risk after	Recommendations
N04	Source node 17 – Ammonia refrigeration: 105-J, unitized chiller 120-C, flash drums 120-CF1…CF4, condenser 127-C, receiver 149-D	9	20	10	R-006, R-007, R-011, R-012, R-013, R-014, R-015, R-016
N05	Source node 18 – Loop purge-gas ammonia recovery and ammonia distillation column 160-C	7	15	10	R-005, R-012, R-017, R-018

5.1 HAZOP worksheets

N01 – Source node 3 – Feed gas and steam to the primary reformer 101-B and secondary reformer 103-D

Ref	Deviation	Causes	Consequences / comments	Safeguards	Risk before	Risk after	Rec.
N01-01	Flow / NO No / low steam flow to the reformer feed	• Steam supply failure (boiler trip, FV-1002 closes) • FIC-1002 transmitter fails high	Steam-to-carbon ratio falls; carbon deposits on the reformer catalyst and tubes overheat – tube rupture, fire in the 101-B firebox	• FSLL-1201 – Low-low steam flow trip FSLL-1201 to the reformer feed trip (source) [IPL] • FFSLL-1201 – Low steam-to-carbon ratio trip FFSLL-1201 closes feed valve XV-1201 (source) [IPL] • TAH-1325 – Reformer tube outlet temperature high alarm TAH-1325 with operator response	4 x 3 = 12	4 x 1 = 4	R-001
N01-02	Flow / MORE High feed gas flow	• FV-1001 fails open • FIC-1001 setpoint error	Low steam-to-carbon ratio (as N01-01) and under-reforming; higher methane slip downstream	• FFSLL-1201 – Low steam-to-carbon ratio trip FFSLL-1201 closes feed valve XV-1201 (source) [IPL] • FIC-1001 – Feed flow control FIC-1001 with ratio station FFI-1001	3 x 3 = 9	3 x 2 = 6	R-001
N01-03	Flow / REVERSE Reverse flow of process gas into the steam header	• Steam pressure lower than reformer pressure (steam trip) • Non-return valve passing	Process gas (H2, CO) into the steam system – flammable and toxic gas in the steam header and at steam users	• Non-return valves on the steam, process-air and feed lines (source) • FSLL-1201 – Low-low steam flow trip FSLL-1201 to the reformer feed trip (source) [IPL]	4 x 2 = 8	4 x 1 = 4	
N01-04	Temperature / MORE High reformer tube temperature	• Over-firing (fuel gas PV-1001B fails open) • Flow maldistribution / plugged tubes • Carbon formation (low steam)	Tube creep and rupture; release of syngas into the firebox – fire, possible firebox overpressure	• TAH-1325 – Reformer tube outlet temperature high alarm TAH-1325 with operator response • Reformer tube-wall temperature surveys (IR pyrometer) and tube creep inspection at each turnaround • PSHH-1221 – High-high fuel gas pressure trip PSHH-1221 (source) [IPL]	4 x 3 = 12	4 x 2 = 8	R-002
N01-05	Pressure / MORE High reformer inlet pressure	• Downstream blockage / 103-D outlet valve closed • Feed compressor over-speed	Reformer tubes and the mixed-feed coil above design	• PSHH-1059 – High-high reformer inlet pressure trip PSHH-1059 (source) [IPL]	4 x 2 = 8	4 x 1 = 4	
N01-06	Flow / MORE High process-air flow to the secondary reformer 103-D	• Process-air control valve fails open • Air compressor surge recovery	Over-temperature in 103-D (combustion zone) – refractory and shell damage; hot spots on the 103-D shell	• FFSHH-1203 – High process-air-to-steam ratio trip FFSHH-1203 closes the process air (source) [IPL] • Non-return valves on the steam, process-air and feed lines (source)	4 x 3 = 12	4 x 2 = 8	R-003
N01-07	Composition / AS WELL AS Sulphur slip from the desulfurizer	• Spent desulfurizer bed • Hydrotreater upset	Poisoning of the reformer catalyst – tube hot bands, reduced conversion	• Reformer tube-wall temperature surveys (IR pyrometer) and tube creep inspection at each turnaround	2 x 3 = 6	2 x 2 = 4	
N01-08	Utilities / services / LOSS OF Loss of instrument air or power	• Instrument-air compressor trip • Power failure	Feed valves fail closed (fail-safe), reformer trips; uncontrolled restart risks low steam-to-carbon	• FFSLL-1201 – Low steam-to-carbon ratio trip FFSLL-1201 closes feed valve XV-1201 (source) [IPL] • Non-return valves on the steam, process-air and feed lines (source)	3 x 3 = 9	3 x 2 = 6	
N01-09	Start-up / REVIEW Start-up – steam before feed sequence	• Feed introduced before steam flow is established	Carbon formation and tube overheating at start-up	• FSLL-1201 – Low-low steam flow trip FSLL-1201 to the reformer feed trip (source) [IPL] • FFSLL-1201 – Low steam-to-carbon ratio trip	4 x 2 = 8	4 x 1 = 4	R-001

Ref	Deviation	Causes	Consequences / comments	Safeguards	Risk before	Risk after	Rec.
				FFSLL-1201 closes feed valve XV-1201 (source) [IPL]			

N02 – Source node 23 – Fuel gas to the primary reformer arch burners 101-BBA, tunnel burners 101-BBT and superheater burners 101-BBS

Ref	Deviation	Causes	Consequences / comments	Safeguards	Risk before	Risk after	Rec.
N02-01	Flow / NO Loss of fuel gas – burner flame-out	• Fuel-gas supply failure • PV-1001B closes • Liquid slug in the fuel gas	Burners go out; if gas returns with the valves open, unburnt gas accumulates in the 101-B firebox – explosion on relight	• PSLL-1221 – Low-low fuel gas pressure trip PSLL-1221 (source) [IPL] • BSLL-1200 – Reformer BMS: flame scanners BSLL-1200 / BSLL-1201 shut the arch-burner valves XV-1220A/C and XV-1222A/C on flame failure [IPL] • Fuel-gas knock-out drum with high-level alarm LAH-1299	5 x 3 = 15	5 x 1 = 5	R-004
N02-02	Pressure / MORE High fuel-gas pressure at the burners	• PV-1001B fails open • Purge gas from N05 at high pressure	Over-firing, flame lift-off and flame-out with gas flowing – firebox explosion; tube overheating	• PSHH-1221 – High-high fuel gas pressure trip PSHH-1221 (source) [IPL] • BSLL-1200 – Reformer BMS: flame scanners BSLL-1200 / BSLL-1201 shut the arch-burner valves XV-1220A/C and XV-1222A/C on flame failure [IPL]	5 x 3 = 15	5 x 1 = 5	R-004
N02-03	Pressure / LESS Low fuel-gas pressure	• Supply drop • Fuel-gas filter blocked	Flame instability, flame-out	• PSLL-1221 – Low-low fuel gas pressure trip PSLL-1221 (source) [IPL]	4 x 3 = 12	4 x 2 = 8	
N02-04	Composition / AS WELL AS Ammonia or liquids in the fuel gas	• Ammonia carry-over in purge gas from N05 • Condensate in the fuel header	NOx emissions, burner fouling and flame instability; liquid can extinguish burners	• Fuel-gas knock-out drum with high-level alarm LAH-1299 • FAL-1064 – Wash-water flow control and low-flow alarm FAL-1064 on the purge-gas scrubber 124-D (source)	3 x 3 = 9	3 x 2 = 6	R-005
N02-05	Utilities / services / LOSS OF Loss of induced draught (ID fan trip)	• ID fan trip • Damper closes	Firebox goes to positive pressure – flames and flue gas out of the peep holes; burns to operators	• PAHH-1226 – Firebox pressure high-high trip PAHH-1226 on loss of induced draught (source) [IPL]	4 x 3 = 12	4 x 2 = 8	
N02-06	Start-up / REVIEW Light-off without adequate purge	• Purge timer bypassed • Fuel valves passing into the cold firebox	Explosive mixture in the firebox at light-off – firebox explosion	• BMS purge interlock – furnace purged with proven air flow before light-off [IPL] • Burner light-off procedure and checklist	5 x 3 = 15	5 x 2 = 10	R-004
N02-07	Level / MORE High liquid level in the fuel-gas knock-out drum	• Condensate accumulation • Drain blocked	Liquid carry-over to the burners (as N02-04)	• Fuel-gas knock-out drum with high-level alarm LAH-1299	3 x 3 = 9	3 x 2 = 6	
N02-08	Fire and explosion / REVIEW Fuel-gas leak at the burner manifold	• Flange / flexible hose failure at a burner • Valve gland leak	Gas fire on the burner platform; operator burns	• Permit-to-work with positive isolation, draining to closed system and gas test before line break • Ammonia PPE: full-face respirators, gas-tight suits for line breaking, eyewash and safety showers	4 x 3 = 12	4 x 2 = 8	

N03 – Source node 15 – Synthesis gas compressor 103-J and ammonia converter 105-D

Ref	Deviation	Causes	Consequences / comments	Safeguards	Risk before	Risk after	Rec.
N03-01	Flow / NO No flow – 103-J trips	• Turbine 103-JT trip • Seal-gas failure (PSLL-1155) • Surge	Loop circulation stops; converter 105-D heats up if the start-up heater 102-B is firing; plant shutdown	• PSLL-1155 – Seal-gas and suction low-low pressure trip PSLL-1155 on 103-J (source) [IPL] • FV-1008 – Anti-surge control FIC-1008 / FV-1008 on 103-J (source)	3 x 3 = 9	3 x 2 = 6	R-020
N03-02	Flow / LESS Low suction flow – compressor surge	• Suction strainer choking (source) • Anti-surge valve FV-1008 fails closed	Surge – thrust-bearing and seal damage, possible seal failure and hydrogen release at 103-J	• FV-1008 – Anti-surge control FIC-1008 / FV-1008 on 103-J (source) • Differential-pressure indication across the 105-J suction strainers (source recommendation)	4 x 3 = 12	4 x 2 = 8	R-006, R-007
N03-03	Temperature / MORE Converter catalyst over-	• Loss of quench / bypass control • High H2 / N2 ratio	Catalyst damage; converter shell and outlet exchanger above design temperature – flange leak	• TSHH-1396 – Converter high-high catalyst temperature trip TSHH-1396: trips 103-J and	5 x 4 = 20	5 x 4 = 20	R-008

Ref	Deviation	Causes	Consequences / comments	Safeguards	Risk before	Risk after	Rec.
	temperature	• Start-up heater 102-B over-firing	or rupture, high-temperature hydrogen attack; syngas fire Quantified in fault tree FT-AM-02 (top 1.8E-2 /yr); threat T1 of bow tie BT-AM-02. Likelihood after safeguards from the calculated frequency 1.8E-2 /yr.	closes the converter feed (source TAH-1383 / TSHH-1396) [IPL] • TAH-1383 – Converter bed temperature alarm TAH-1383 with operator response [IPL] • Loop equipment designed for 160 barg; materials selected per the Nelson curves (hydrogen service)			
N03-04	Pressure / MORE High synthesis loop pressure	• Blocked outlet / loss of refrigeration (N04) • High inerts – purge closed • 103-J over-speed	Loop above design (160 barg) – flange failure, large syngas release	• PSV-1018 – Synthesis loop relief valve PSV-1018 at 155 barg to the flare [IPL] • AT-1000 – Loop gas analyser AT-1000 (H2 / N2 ratio, inerts) with alarms • HV-1019 – Remote depressurisation of the synthesis loop to the flare via HV-1019	5 x 3 = 15	5 x 1 = 5	R-009
N03-05	Composition / AS WELL AS High inerts / wrong H2:N2 ratio	• Purge flow low (N05) • Front-end upset	Loop pressure rises; conversion drops; temperature excursions in 105-D	• AT-1000 – Loop gas analyser AT-1000 (H2 / N2 ratio, inerts) with alarms	3 x 3 = 9	3 x 2 = 6	
N03-06	Corrosion / erosion / MORE High-temperature hydrogen attack of loop exchangers	• Operation above the Nelson-curve limit • Material mix-up	Sudden failure of 123-C1 / 124-C1 – syngas jet fire	• Loop equipment designed for 160 barg; materials selected per the Nelson curves (hydrogen service) [IPL]	5 x 2 = 10	5 x 1 = 5	R-010
N03-07	Flow / REVERSE Reverse rotation / back-flow on compressor trip	• Discharge check valve fails to close	Compressor reverse rotation – seal and bearing damage, gas release at the seals	• Non-return valves on the steam, process-air and feed lines (source) • PSLL-1155 – Seal-gas and suction low-low pressure trip PSLL-1155 on 103-J (source) [IPL]	4 x 2 = 8	4 x 1 = 4	
N03-08	Fire and explosion / REVIEW Syngas leak at the compressor or loop flanges	• Seal failure • Flange leak on hot hydrogen service	Hydrogen-rich jet fire; vapour cloud explosion in the congested compressor house	• UCP-1000 – Unit ESD and emergency depressurisation (logic UCP-1000) [IPL] • HV-1019 – Remote depressurisation of the synthesis loop to the flare via HV-1019	5 x 3 = 15	5 x 2 = 10	R-019
N03-09	Start-up / REVIEW Start-up heater 102-B firing without loop circulation	• 102-B lit before 103-J runs • 103-J trips during heat-up	Converter overheating at start-up	• TSHH-1396 – Converter high-high catalyst temperature trip TSHH-1396: trips 103-J and closes the converter feed (source TAH-1383 / TSHH-1396) [IPL]	4 x 3 = 12	4 x 2 = 8	R-020

N04 – Source node 17 – Ammonia refrigeration: 105-J, unitized chiller 120-C, flash drums 120-CF1…CF4, condenser 127-C, receiver 149-D

Ref	Deviation	Causes	Consequences / comments	Safeguards	Risk before	Risk after	Rec.
N04-01	Pressure / MORE High pressure in the refrigerant condenser 127-C and receiver 149-D	• Loss of cooling water to 127-C • Non-condensables accumulation • Discharge valve closed	Refrigerant system above design; PSV-1109 lifts to the flare; if relief fails – rupture and large liquid ammonia release (toxic cloud) Quantified in fault tree FT-AM-01 (top 2.4E-6 /yr); threat T1 of bow tie BT-AM-01. Likelihood after safeguards from the calculated frequency 2.4E-6 /yr.	• PAH-1009 – Refrigerant pressure alarm PAH-1009 / PAH-1109 with operator response (source) [IPL] • PSHH-1110 – 105-J discharge high-high pressure trip PSHH-1110 [IPL] • PSV-1109 – Receiver 149-D relief valve PSV-1109 at 20 barg to the flare [IPL] • Cooling-water low-flow alarm at condenser 127-C	5 x 4 = 20	5 x 1 = 5	R-011, R-015
N04-02	Level / MORE High level in a flash drum 120-CF1…CF4	• LV-1015A fails open • Load change	Liquid ammonia carried into the 105-J suction – compressor damage, seal failure and ammonia release	• LSHH-1214 – High-high level trips LSHH-1214…1217 on the flash drums 120-CF1…CF4 trip 105-J (source) [IPL] • LIC-1015A – Flash-drum level control LIC-1015A / LV-1015A with LAH / LAL alarms (source)	4 x 3 = 12	4 x 2 = 8	
N04-03	Pressure / LESS Low pressure / vacuum in flash drums 120-CF1 / 120-CF2	• No / low vapour flow (source) • Loss of boil-off from storage – XV-1801 closes	Loss of suction to the 105-J stages – compressor damage; vacuum in the flash drums (designed for 0.25 bar vacuum)	• 120-CF1 – Flash drums 120-CF1/CF2 designed for 0.25 bar vacuum (source) [IPL] • FV-1008 – Anti-surge control FIC-1008 / FV-1008 on 103-J (source)	3 x 3 = 9	3 x 2 = 6	
N04-04	Flow / LESS Choking of the 105-J suction strainers	• Strainer fouling (source)	Loss of suction to a stage of 105-J – surge and compressor damage	• Differential-pressure indication across the 105-J suction strainers (source recommendation)	3 x 3 = 9	3 x 2 = 6	R-006, R-007
N04-05	Temperature / LESS Low temperature – auto-	• Rapid depressurisation of a liquid-full section	Brittle fracture of carbon steel below −33 °C – sudden ammonia release	• Refrigeration circuit in low-temperature carbon steel rated to −33 °C [IPL]	5 x 2 = 10	5 x 1 = 5	R-014

Ref	Deviation	Causes	Consequences / comments	Safeguards	Risk before	Risk after	Rec.
	refrigeration on depressurisation	• Ammonia flashing through a passing valve					
N04-06	Utilities / services / LOSS OF Loss of cooling water	• Cooling-water pump trip • Cooling-tower failure	Condenser 127-C pressure rises (as N04-01)	• Cooling-water low-flow alarm at condenser 127-C • PSHH-1110 – 105-J discharge high-high pressure trip PSHH-1110 [IPL]	4 x 3 = 12	4 x 2 = 8	R-015
N04-07	Toxicity / REVIEW Ammonia leak from the refrigerant pumps, flanges or receiver	• Pump seal failure • Flange gasket failure • Small-bore fitting fatigue	Toxic ammonia cloud in the refrigeration area; exposure of operators and, for large releases, people off site	• AT-1801A – Ammonia detectors AT-1801A/B/C (1oo3) at the refrigeration area and pump bay with ESD of XV-1801 and the receiver outlet XV-1290 (logic UCP-1000) [IPL] • Fixed water-spray curtain around the receiver 149-D and the refrigerant pumps • Toxic-gas refuge (control room positive pressure, HVAC trip on ammonia), escape sets and wind socks • Ammonia PPE: full-face respirators, gas-tight suits for line breaking, eyewash and safety showers	5 x 3 = 15	5 x 2 = 10	R-012, R-013, R-016
N04-08	Maintenance / INADEQUATE Line break on ammonia service	• Incomplete draining / isolation • Trapped liquid between valves	Ammonia spray onto the maintenance crew – chemical burns, inhalation	• Permit-to-work with positive isolation, draining to closed system and gas test before line break • Ammonia PPE: full-face respirators, gas-tight suits for line breaking, eyewash and safety showers	4 x 3 = 12	4 x 2 = 8	
N04-09	Relief / INADEQUATE Relief discharge of ammonia to the flare	• PSV-1109 lifts on overpressure	Ammonia to the flare – NOx; if the flare is out, toxic ammonia release at height	• PSV-1109 – Receiver 149-D relief valve PSV-1109 at 20 barg to the flare	3 x 3 = 9	3 x 2 = 6	

N05 – Source node 18 – Loop purge-gas ammonia recovery and ammonia distillation column 160-C

Ref	Deviation	Causes	Consequences / comments	Safeguards	Risk before	Risk after	Rec.
N05-01	Flow / NO No wash-water flow to the purge-gas scrubber 124-D	• Wash-water pump trip • FV-1029 closes	Ammonia slips into the purge gas to fuel (N02) – NOx and burner problems; ammonia odour around the reformer	• FAL-1064 – Wash-water flow control and low-flow alarm FAL-1064 on the purge-gas scrubber 124-D (source)	3 x 3 = 9	3 x 2 = 6	R-005
N05-02	Pressure / MORE High pressure in the distillation column 160-C	• Purge let-down PV-1033A fails open – gas blow-by from 140 barg • Reboiler 161-C1 over-heating	Column (20 barg design) over-pressured – rupture and ammonia release	• PSV-1033 – Column 160-C relief valve PSV-1033 at 20 barg [IPL] • PV-1033A – Purge let-down valve PV-1033A with downstream relief sized for gas blow-by [IPL] • PIC-1033 – Column pressure control PIC-1033 with PAH-1033 (source)	5 x 3 = 15	5 x 2 = 10	R-018
N05-03	Level / MORE High level in column 160-C	• LV-1049 fails closed • Reflux pump 161-J trip	Liquid ammonia carry-over into the purge gas to fuel	• PIC-1033 – Column pressure control PIC-1033 with PAH-1033 (source)	3 x 3 = 9	3 x 2 = 6	R-017
N05-04	Level / LESS Low level in the purge-gas scrubber 124-D	• LV-1027 fails open	Gas blow-by from 140 barg into the low-pressure ammonia return (as N05-02)	• PV-1033A – Purge let-down valve PV-1033A with downstream relief sized for gas blow-by [IPL] • PSV-1033 – Column 160-C relief valve PSV-1033 at 20 barg [IPL]	5 x 2 = 10	5 x 1 = 5	R-018
N05-05	Temperature / MORE High column bottoms temperature	• Reboiler steam control fails open	Column pressure rises; water carry-over into the recovered ammonia	• PIC-1033 – Column pressure control PIC-1033 with PAH-1033 (source) • PSV-1033 – Column 160-C relief valve PSV-1033 at 20 barg [IPL]	3 x 3 = 9	3 x 2 = 6	
N05-06	Toxicity / REVIEW Ammonia leak at the column pumps 161-J	• Seal failure • Flange leak	Toxic ammonia release at the recovery unit	• AT-1801A – Ammonia detectors AT-1801A/B/C (1oo3) at the refrigeration area and pump bay with ESD of XV-1801 and the receiver outlet XV-1290 (logic UCP-1000) • Ammonia PPE: full-face respirators, gas-tight suits for line breaking, eyewash and safety showers	4 x 3 = 12	4 x 2 = 8	R-012
N05-07	Utilities / services / LOSS OF Power failure	• Grid failure	Pumps stop, column pressure falls; purge flow to fuel stops – reformer fuel balance upset	• UCP-1000 – Unit ESD and emergency depressurisation (logic UCP-1000)	3 x 3 = 9	3 x 2 = 6	

5.2 Layer of protection analysis (LOPA)

Study LP-EX-AM-001 (LOPA Study Manager), 2026-11-20. The scenarios are the HAZOP deviations whose IPLs were ticked LOPA – one scenario for each initiating cause, as the tool imports them. Explicit LOPA: initiating-event frequency × enabling conditions × conditional modifiers × IPL PFDs, compared with the target maximum event likelihood (TMEL) for each receptor. Only IPL credits accepted in the adjudication are counted (seven tests: specific, independent, dependable, auditable, access & change control, in time, cyber-secure). Targets are the typical example values of the tool (safety 1E-6 /yr at severity 5, 1E-5 at 4).

Eleven scenarios from four HAZOP deviations. The two operator alarms are rejected because each shares its sensor with the trip it backs up – the same common causes the fault trees model (FT-AM-01 BE-009, FT-AM-02 BE-010) – and the remaining gaps point to separate sensors (R-011, R-008), a double block and bleed and dry gas seals.

LOPA	HAZOP	Initiating cause	IE /yr	IPLs (PFD); + = proposed	Result – worst receptor	With proposed IPLs
L-01	N04-01	Loss of cooling water to 127-C	0.1	PSHH-1110 0.01 PAH-1009 0.1 – rejected PSV-1109 0.01 + PAH-1009 on its own transmit 0.1	SIL 1 needed (Community)	Gap ≤ 10× – one more IPL or SIL 1
L-02	N04-01	Non-condensables accumulation	0.1	PSHH-1110 0.01 PAH-1009 0.1 – rejected PSV-1109 0.01 + PAH-1009 on its own transmit 0.1	SIL 1 needed (Community)	Gap ≤ 10× – one more IPL or SIL 1
L-03	N04-01	Discharge valve closed	0.02	PSHH-1110 0.01 PAH-1009 0.1 – rejected PSV-1109 0.01 + PAH-1009 on its own transmit 0.1	Gap ≤ 10× – one more IPL or SIL 1 (Community)	Meets target
L-04	N03-03	Loss of quench / bypass control	0.1	TSHH-1396 0.01 TAH-1383 0.1 – rejected + Second converter trip on a s 0.01	SIL 2 needed (Safety)	Gap ≤ 10× – one more IPL or SIL 1
L-05	N03-03	High H2 / N2 ratio	0.2	TSHH-1396 0.01 TAH-1383 0.1 – rejected + Second converter trip on a s 0.01	SIL 2 needed (Safety)	Gap ≤ 10× – one more IPL or SIL 1
L-06	N03-03	Start-up heater 102-B over-firing	0.05	TSHH-1396 0.01 TAH-1383 0.1 – rejected + Second converter trip on a s 0.01	SIL 1 needed (Safety)	Meets target
L-07	N02-01	Fuel-gas supply failure	0.1	PSLL-1221 0.1 BSLL-1200 0.01 + Fuel-gas double block and bl 0.1	Gap ≤ 10× – one more IPL or SIL 1 (Safety)	Meets target
L-08	N02-01	PV-1001B closes	0.1	PSLL-1221 0.1 BSLL-1200 0.01 + Fuel-gas double block and bl 0.1	Gap ≤ 10× – one more IPL or SIL 1 (Safety)	Meets target
L-09	N02-01	Liquid slug in the fuel gas	0.05	PSLL-1221 0.1 BSLL-1200 0.01 + Fuel-gas double block and bl 0.1	Gap ≤ 10× – one more IPL or SIL 1 (Safety)	Meets target
L-10	N03-08	Seal failure	0.1	UCP-1000 0.1 + Dry gas seals with seal-gas 0.1	SIL 1 needed (Safety)	Gap ≤ 10× – one more IPL or SIL 1
L-11	N03-08	Flange leak on hot hydrogen service	0.01	UCP-1000 0.1 + Dry gas seals with seal-gas 0.1	Gap ≤ 10× – one more IPL or SIL 1 (Safety)	Meets target

IPL credits rejected in the adjudication:

Scenarios	IPL	Failed test	Reason
L-01, L-02, L-03	PAH-1009 – Refrigerant pressure alarm PAH-1009 / PAH-1109 with operator response (source)	Independent	Not independent of the PSHH-1110 trip – one transmitter blinds both (HAZOP R-011)
L-04, L-05, L-06	TAH-1383 – Converter bed temperature alarm TAH-1383 with operator response	Independent	Not independent of the TSHH-1396 trip – one thermocouple blinds both (HAZOP R-008)

SCAI register – safety controls, alarms and interlocks credited, with the SIL each needs (from the PFD credited):

Tag	Function	PFD credited	SIL needed	Scenarios	Proof test	Owner
BSLL-1200 –	BSLL-1200 – Reformer BMS: flame scanners BSLL-1200 / BSLL-1201 shut the arch-burner valves XV-1220A/C and XV-1222A/C on flame failure	0.01	SIL 2	L-07, L-08, L-09	6 months	E. Instrument
	Dry gas seals with seal-gas pressure trip (recommendation)	0.1	SIL 1	L-10, L-11		
	Fuel-gas double block and bleed, interlocked with the pre-ignition purge (recommendation)	0.1	SIL 1	L-07, L-08, L-09		
	PAH-1009 on its own transmitter, separate from PSHH-1110 (R-011)	0.1	Not a SIF (credit ≤ RRF 10)	L-01, L-02, L-03		
PSHH-1110 –	PSHH-1110 – 105-J discharge high-high pressure trip PSHH-1110	0.01	SIL 1	L-01, L-02, L-03	12 months	E. Instrument
PSLL-1221 –	PSLL-1221 – Low-low fuel gas pressure trip PSLL-1221 (source)	0.1	SIL 1	L-07, L-08, L-09	12 months	E. Instrument
	Second converter trip on a separate bed thermocouple, SIL 2 (R-008)	0.01	SIL 2	L-04, L-05, L-06		
TSHH-1396 –	TSHH-1396 – Converter high-high catalyst temperature trip TSHH-1396: trips 103-J and closes the converter feed (source TAH-1383 / TSHH-1396)	0.01	SIL 1	L-04, L-05, L-06	12 months	E. Instrument
UCP-1000 –	UCP-1000 – Unit ESD and emergency depressurisation (logic UCP-1000)	0.1	SIL 1	L-10, L-11	12 months	E. Instrument

A LOPA that credited PAH-1009 and PSHH-1110 as two independent layers would take 0.101 × 0.0102 = 1.0E-3; FT-AM-01 shows the pair fails with 0.051 because one transmitter serves both. The adjudication catches the same dependency with the independence test, and the proposed separate transmitter (R-011) restores the alarm credit.

PSV-1109 is credited at the CCPS LOPA screening value 0.01, while the fault tree uses the equipment-data mean 2.1E-4: LOPA screening values are deliberately conservative.

6 FMECA results

Systems S01–S05 correspond one-to-one with the HAZOP nodes. RPN = S x O x D (1–10 scales); action priority from the AIAG-VDA table in the workbook.

Ref	Tag	Component	Failure mode	End effect	S·O·D	RPN	Recommended action	After	HAZOP
S01-01	FFSLL-1201	Transmitter electronics	FTF – Fail to function on demand	Tube overheating and rupture	9·3·6	162	Separate transmitters from FFI-1001 (R-001)	9·2·4 = 72	HAZOP N01-01
S01-02	101-B	Burner nozzle	PDE – Incorrect air / fuel ratio	Tube rupture – fire	8·4·4	128	Fixed IR tube-wall monitoring (R-002)	8·3·2 = 48	HAZOP N01-04
S01-03	XV-1201	Actuator	FTC – Fail to close on demand	Carbon formation / overheating	9·3·6	162	–	–	HAZOP N01-01
S02-01	BSLL-1200	Sensing element	FTF – Fail to function on demand	Firebox explosion	10·3·6	180	BMS verification to IEC 61511 (R-004)	10·2·4 = 80	HAZOP N02-01
S02-02	XV-1220A	Seats	LCP – Leakage in closed position	Explosion at light-off	10·3·6	180	–	–	HAZOP N02-06
S02-03	PV-1001B	Trim (plug / seat)	FTC – Fail to close on demand	Flame lift-off / tube overheating	8·4·3	96	–	–	HAZOP N02-02
S03-01	TSHH-1396	Sensing element	FTF – Fail to function on demand	Converter / exchanger failure – syngas fire	10·3·6	180	Second independent thermocouple, 1oo2 (R-008)	10·2·4 = 80	HAZOP N03-03 Fault tree FT-AM-02 event BE-006
S03-02	103-J	Dry gas / oil seals	ELP – External leakage – process medium	Jet fire / VCE (BT-AM-02 T2)	10·3·4	120	–	–	HAZOP N03-02
S03-03	123-C1	Tubes	STD – Structural deficiency	Jet fire (BT-AM-02 T3)	10·2·6	120	HTHA review (R-010)	10·2·4 = 80	HAZOP N03-06
S03-04	XV-1250A	Actuator	FTC – Fail to close on demand	Over-temperature continues	9·3·6	162	–	–	HAZOP N03-03 Fault tree FT-AM-02 event BE-008
S04-01	PSHH-1110	Transmitter electronics	FTF – Fail to function on demand	PSV-1109 is the last layer (FT-AM-01)	9·3·6	162	Independent transmitter (R-011)	9·2·4 = 72	HAZOP N04-01 Fault tree FT-AM-01 event BE-006
S04-02	PSV-1109	Nozzle and disc	FTO – Fail to open on	Rupture – toxic ammonia	10·2·6	120	–	–	HAZOP N04-01

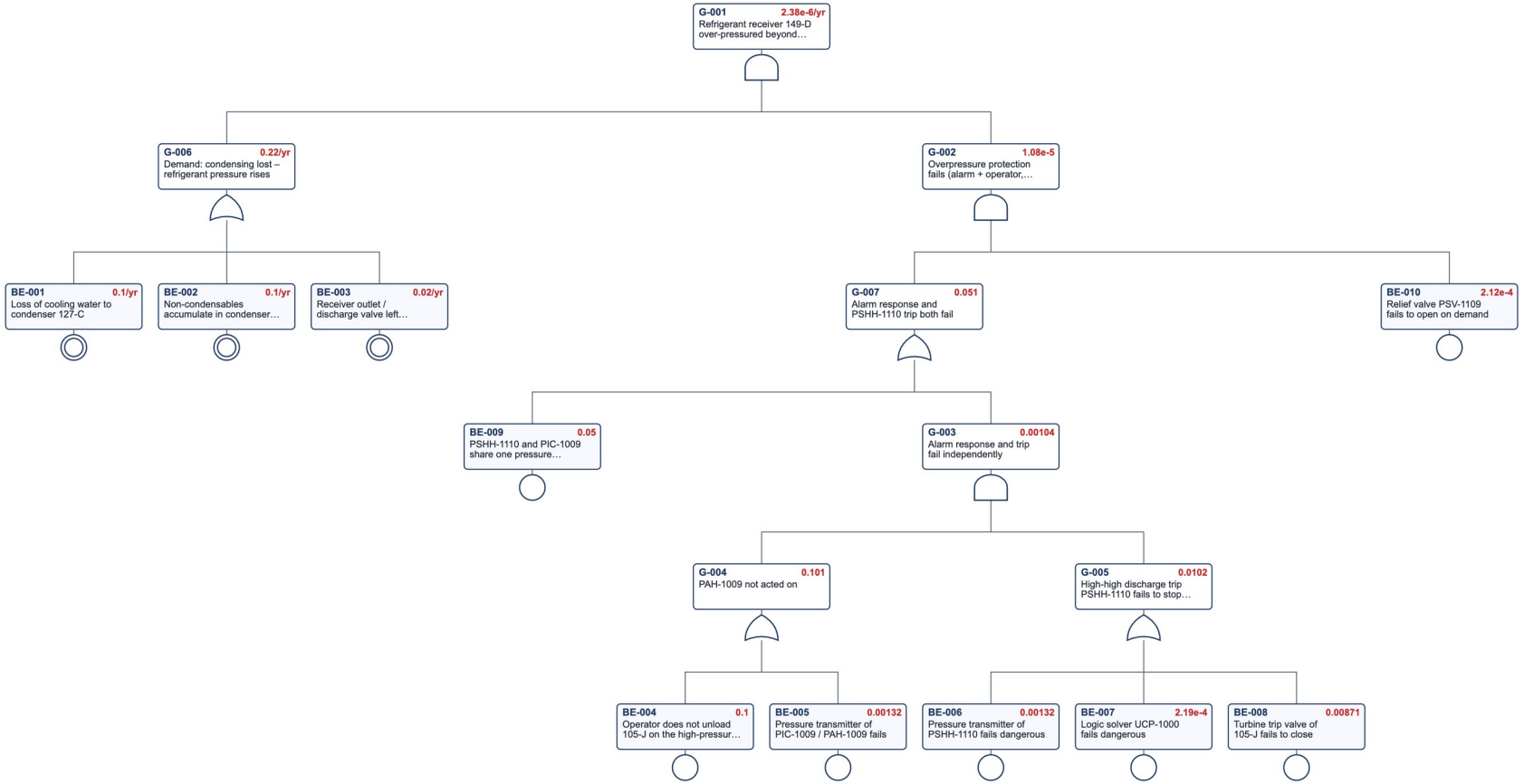
Ref	Tag	Component	Failure mode	End effect	S·O·D	RPN	Recommended action	After	HAZOP
			demand	release					Fault tree FT-AM-01 event BE-010
S04-03	127-C	Tubes	PDE – Reduced heat transfer	FT-AM-01 demand	6·4·3	72	Low-low cooling-water flow trip (R-015)	6·3·3 = 54	HAZOP N04-06
S04-04	AT-1801A	Sensor	FTF – Fail to function on demand	Larger toxic cloud (FT-AM-03)	8·4·4	128	Detector mapping (R-012)	8·3·4 = 96	HAZOP N04-07
S04-05	LSHH-1214	Sensing element	FTF – Fail to function on demand	Compressor damage, seal leak	8·3·6	144	–	–	HAZOP N04-02
S05-01	PV-1033A	Trim (plug / seat)	FTC – Fail to close on demand	Rupture – toxic release	10·3·3	90	Confirm PSV-1033 sizing for blow-by (R-018)	10·2·3 = 60	HAZOP N05-02
S05-02	161-J	Mechanical seal	ELP – External leakage – process medium	Toxic exposure	8·4·4	128	–	–	HAZOP N05-06

7 Fault tree analysis

Every fault tree was calculated by the Fault Tree & PFD Calculator (exact BDD, time-averaged PFDavg over the proof-test cycle). Basic-event values come from the data library where an item fits: 'checked' items were compared with the reference table named; 'typical' items must be verified against the edition used.

7.1 FT-AM-01 – Ammonia refrigeration receiver over-pressured beyond design

Top event: condenser 127-C / receiver 149-D pressure exceeds design because condensing is lost (demand gate G-006) and the alarm PAH-1009 with operator response, the discharge trip PSHH-1110 and relief valve PSV-1109 all fail (gate G-002). PSHH-1110 shares its transmitter with PIC-1009 in the design reviewed (BE-009, R-011): one fault blinds the alarm and the trip, but not PSV-1109. Gate G-002 is the barrier of bow tie BT-AM-01 threat T1. HAZOP N04-01; FMECA S04.



FT-AM-01 as drawn by the Fault Tree & PFD Calculator (red: time-averaged values).

Top-event frequency 2.4E-6 /yr. Gate values: G-001 2.4E-6; G-002 1.1E-5; G-003 1.0E-3; G-004 1.0E-1; G-005 1.0E-2; G-006 2.2E-1; G-007 5.1E-2.

Event	Description	Fussell-Vesely	RAW
BE-010	Relief valve PSV-1109 fails to open on demand	1.000	4.7E3
BE-009	PSHH-1110 and PIC-1009 share one pressure transmitter – both blind	0.980	2.0E1
BE-001	Loss of cooling water to condenser 127-C	0.455	–
BE-002	Non-condensables accumulate in condenser 127-C	0.455	–
BE-003	Receiver outlet / discharge valve left closed after maintenance	0.091	–
BE-004	Operator does not unload 105-J on the high-pressure alarm PAH-1009	0.019	1.2E0

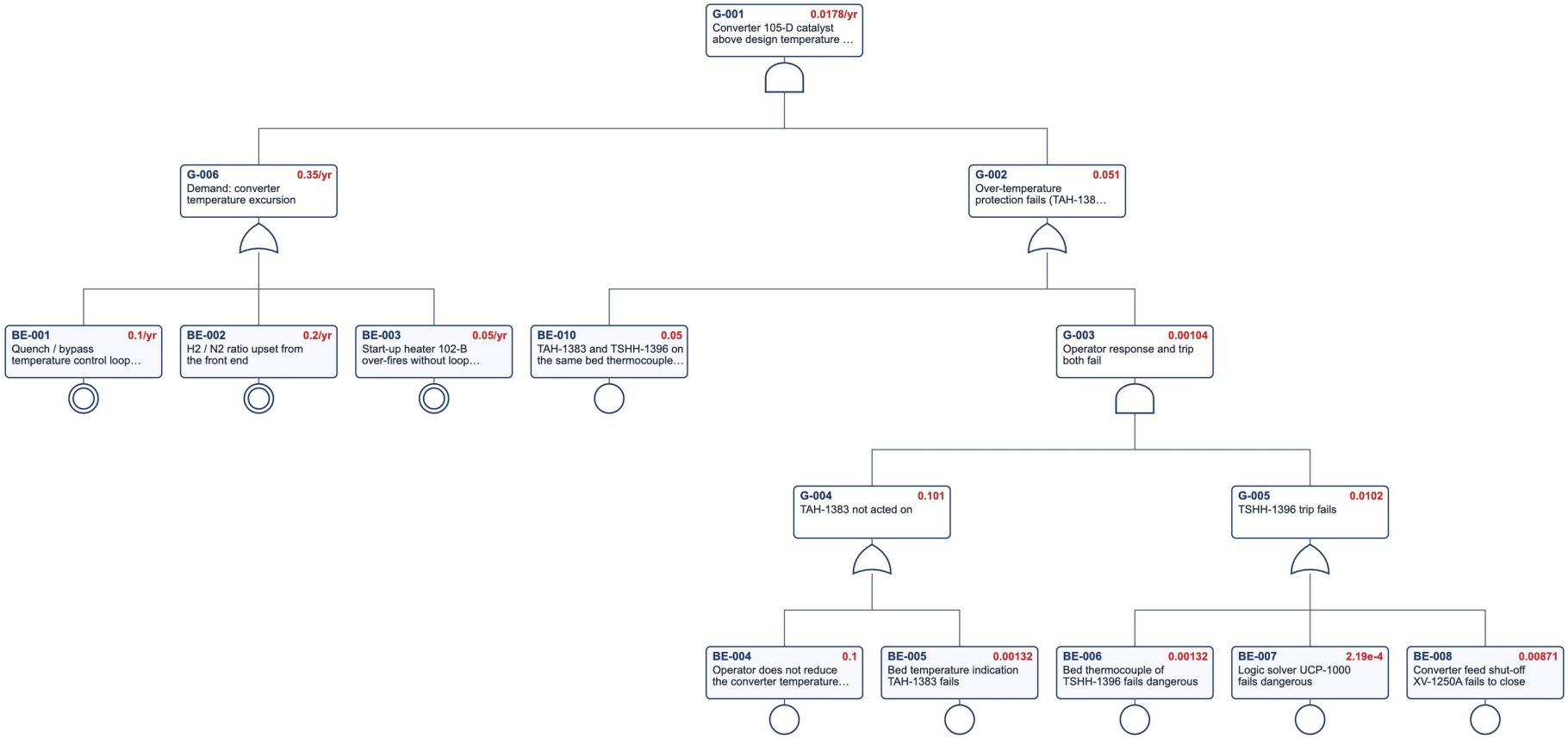
Event	Library item	Value	Status
BE-001	Loss of cooling water	1.0E-1 /yr	checked – CCPS LOPA (2001), Table 5.1
BE-004	Operator response to an independent alarm, ≥ 10 min, simple documented action, clear indication	1.0E-1	checked – CCPS LOPA (2001), Table 6.5; CCPS Guidelines for Hazard Evaluation Procedures, 3rd ed., Table 7.4; CCPS Guidelines for Initiating

Event	Library item	Value	Status
			Events and IPLs in LOPA (2015), Data Table 5.46
BE-005	Pressure transmitter (SIL-rated, smart) – λDU	3.0E-7 /h	typical – verify
BE-006	Pressure transmitter (SIL-rated, smart) – λDU	3.0E-7 /h	typical – verify
BE-007	Safety PLC, per function incl. I/O (certified) – λDU	5.0E-8 /h	typical – verify
BE-008	Shutdown valve (ball / gate) with actuator – λDU	2.0E-6 /h	typical – verify
BE-010	Spring-loaded relief valve fails to open on demand	2.1E-4	checked – CCPS Guidelines for Process Equipment Reliability Data (1989), taxonomy 4.3.3.2 – mean

The protection PFDavg of 1.1E-5 rests on PSV-1109: the shared PIC-1009 transmitter (BE-009) blinds both the alarm and PSHH-1110, so the instrumented layers add little. R-011 gives the trip its own transmitter, so a demand is stopped before the plant relieves ammonia to flare.

7.2 FT-AM-02 – Uncontrolled over-temperature of the ammonia converter 105-D

Top event: the catalyst temperature of 105-D rises above the design limit (demand gate G-006) and both the operator response to TAH-1383 and the high-high trip TSHH-1396 (closes converter feed XV-1250A and trips 103-J) fail (gate G-002). As reviewed, the alarm and the trip use the same bed thermocouple (BE-010, R-008). Gate G-002 is the barrier of bow tie BT-AM-02 threat T1. HAZOP N03-03; FMECA S03.



FT-AM-02 as drawn by the Fault Tree & PFD Calculator (red: time-averaged values).

Top-event frequency 1.8E-2 /yr. Gate values: G-001 1.8E-2; G-002 5.1E-2; G-003 1.0E-3; G-004 1.0E-1; G-005 1.0E-2; G-006 3.5E-1.

Event	Description	Fussell-Vesely	RAW
BE-010	TAH-1383 and TSHH-1396 on the same bed thermocouple – both blind	0.980	2.0E1
BE-002	H2 / N2 ratio upset from the front end	0.571	–
BE-001	Quench / bypass temperature control loop fails	0.286	–
BE-003	Start-up heater 102-B over-fires without loop circulation	0.143	–
BE-004	Operator does not reduce the converter temperature on TAH-1383	0.019	1.2E0
BE-008	Converter feed shut-off XV-1250A fails to close	0.016	2.9E0

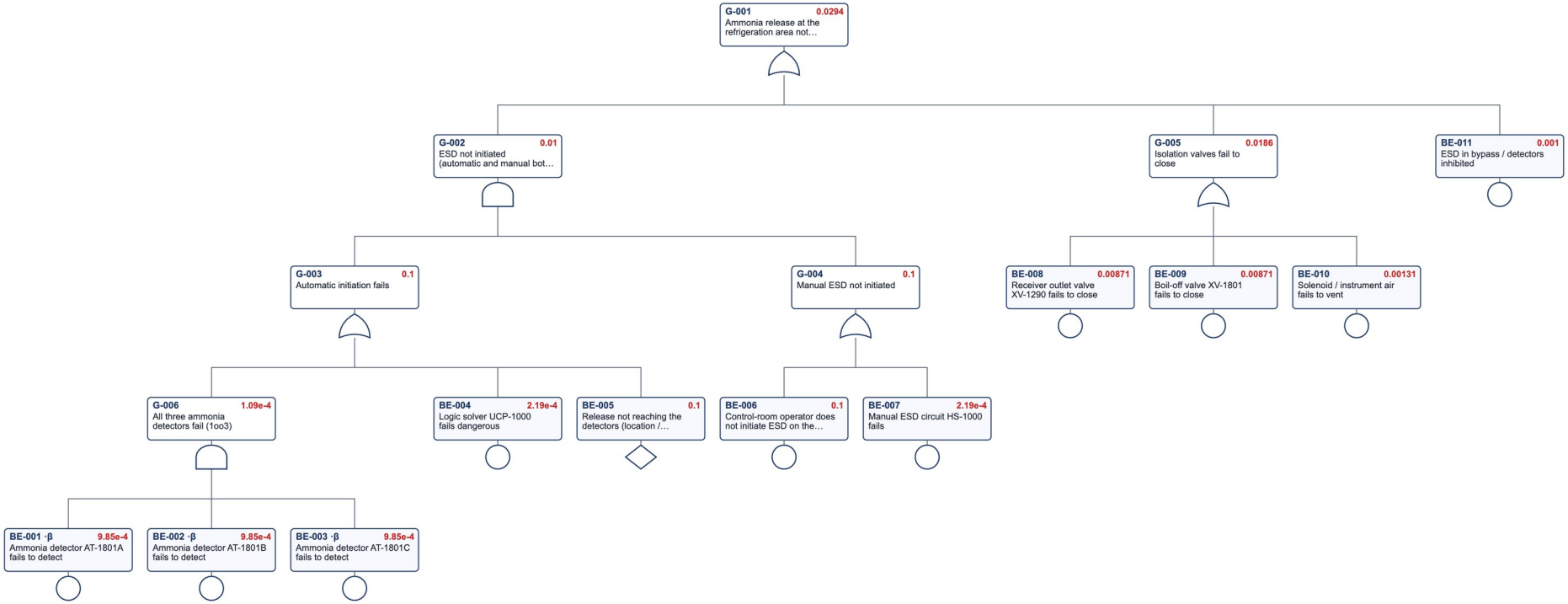
Event	Library item	Value	Status
BE-001	BPCS instrument loop failure	1.0E-1 /yr	checked – CCPS LOPA (2001), Table 5.1; CCPS Guidelines for Initiating Events and IPLs in LOPA (2015), Data Table 4.1
BE-004	Operator response to an independent alarm, ≥ 10 min,	1.0E-1	checked – CCPS LOPA (2001), Table 6.5; CCPS

Event	Library item	Value	Status
	simple documented action, clear indication		Guidelines for Hazard Evaluation Procedures, 3rd ed., Table 7.4; CCPS Guidelines for Initiating Events and IPLs in LOPA (2015), Data Table 5.46
BE-005	Temperature transmitter with element – λDU	3.0E-7 /h	typical – verify
BE-006	Temperature transmitter with element – λDU	3.0E-7 /h	typical – verify
BE-007	Safety PLC, per function incl. I/O (certified) – λDU	5.0E-8 /h	typical – verify
BE-008	Shutdown valve (ball / gate) with actuator – λDU	2.0E-6 /h	typical – verify

The single thermocouple behind TAH-1383 and TSHH-1396 dominates; with R-008 the alarm and the trip become independent layers.

7.3 FT-AM-03 – Ammonia release at the refrigeration area not isolated

Top event (gate G-001): an ammonia release at the receiver 149-D or the refrigerant pumps is not isolated – neither detectors AT-1801A/B/C (1oo3) through UCP-1000 nor the control-room operator (HS-1000) initiate, or XV-1290 / XV-1801 fail to close. PFDavg of G-001 is heading H1 of event tree ETA-AM-01 and the ESD barrier of bow tie BT-AM-01. HAZOP N04-07, N05-06; FMECA S04.



FT-AM-03 as drawn by the Fault Tree & PFD Calculator (red: time-averaged values).

Top-event PFDavg 2.9E-2. Gate values: G-001 2.9E-2; G-002 1.0E-2; G-003 1.0E-1; G-004 1.0E-1; G-005 1.9E-2; G-006 1.1E-4.

Event	Description	Fussell-Vesely	RAW
BE-006	Control-room operator does not initiate ESD on the ammonia alarm	0.334	4.0E0
BE-005	Release not reaching the detectors (location / wind)	0.333	4.0E0
BE-008	Receiver outlet valve XV-1290 fails to close	0.289	3.4E1
BE-009	Boil-off valve XV-1801 fails to close	0.289	3.4E1
BE-010	Solenoid / instrument air fails to vent	0.043	3.4E1
BE-011	ESD in bypass / detectors inhibited	0.033	3.4E1

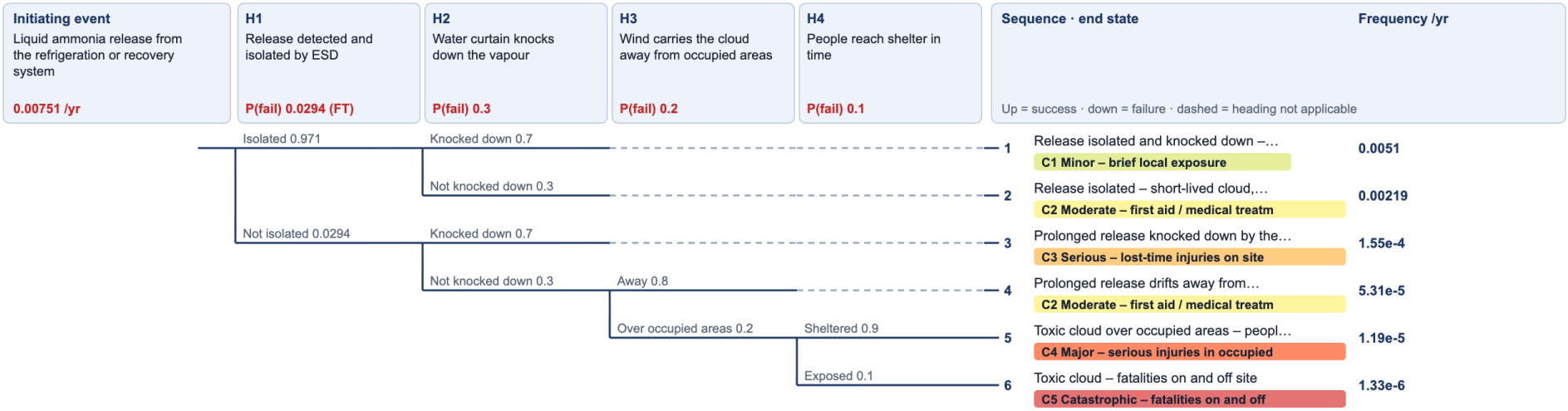
Event	Library item	Value	Status
BE-001	Point gas detector, catalytic – λDU	1.0E-6 /h	typical – verify
BE-002	Point gas detector, catalytic – λDU	1.0E-6 /h	typical – verify
BE-003	Point gas detector, catalytic – λDU	1.0E-6 /h	typical – verify
BE-004	Safety PLC, per function incl. I/O (certified) – λDU	5.0E-8 /h	typical – verify
BE-006	Operator response to an independent alarm, ≥ 10 min, simple documented action, clear indication	1.0E-1	checked – CCPS LOPA (2001), Table 6.5; CCPS Guidelines for Hazard Evaluation Procedures, 3rd ed., Table 7.4; CCPS Guidelines for Initiating Events and IPLs in LOPA (2015), Data Table 5.46
BE-007	Hard-wired relay logic, per function – λDU	5.0E-8 /h	typical – verify
BE-008	Shutdown valve (ball / gate) with actuator – λDU	2.0E-6 /h	typical – verify
BE-009	Shutdown valve (ball / gate) with actuator – λDU	2.0E-6 /h	typical – verify
BE-010	Solenoid / pilot valve on an actuator – λDU	3.0E-7 /h	typical – verify

Detector coverage and the operator's manual ESD dominate the PFDavg of 2.9E-2; R-012 (detector mapping) addresses the coverage.

8 Event tree analysis

8.1 ETA-AM-01 – Liquid ammonia release from the refrigeration system

Initiating event = top event of bow tie BT-AM-01, frequency 7.5E-3 /yr. H1 is linked to fault tree FT-AM-03 (ammonia detection and ESD). End states TOXON, TOXOCC and TOXOFF are reproduced by the bow-tie consequences. HAZOP N04-07. Illustrative probabilities.



ETA-AM-01 as drawn by Event Tree Analysis.

Heading	Success / failure	P(failure)	Source
H1 Release detected and isolated by ESD	Isolated / Not isolated	2.9E-2	PFDavg from fault tree FT-AM-03 gate G-001 = BT-AM-01 ESD barrier
H2 Water curtain knocks down the vapour	Knocked down / Not knocked down	3.0E-1	= BT-AM-01 barrier 'Water-spray curtain'
H3 Wind carries the cloud away from occupied areas	Away / Over occupied areas	2.0E-1	Wind-rose fraction towards the control room, offices and the boundary – illustrative
H4 People reach shelter in time	Sheltered / Exposed	1.0E-1	= BT-AM-01 barrier 'Toxic-gas refuge' (R-016)

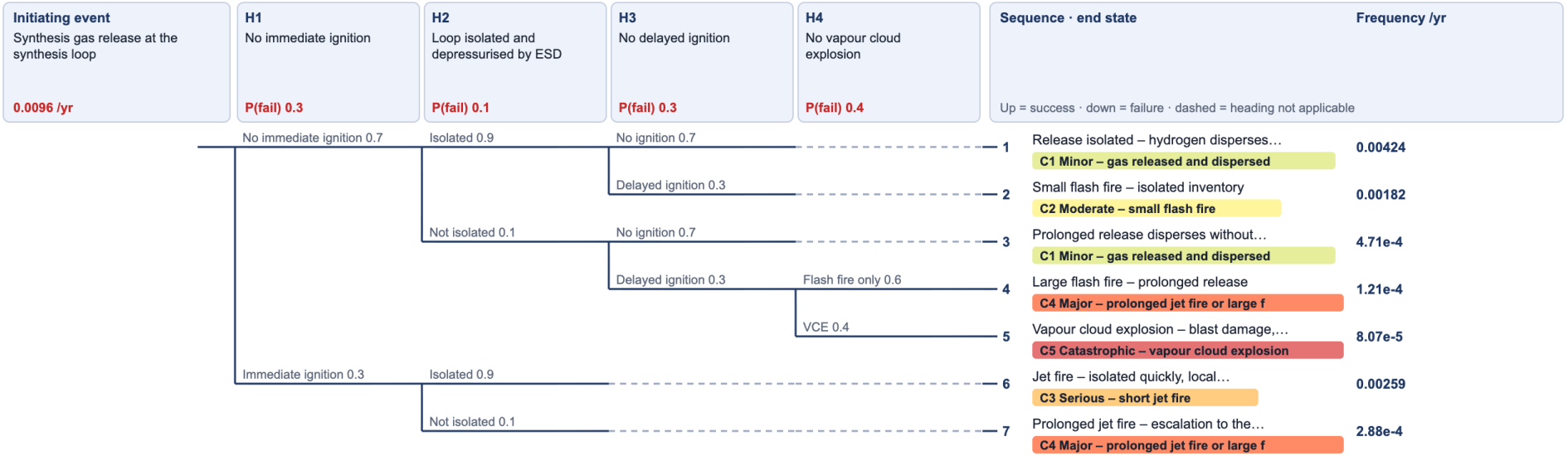
End state	Category	Frequency /yr
Release isolated and knocked down – minor exposure	C1 – Minor – brief local exposure	5.1E-3
Release isolated – short-lived cloud, on-site exposure	C2 – Moderate – first aid / medical treatment on site	2.2E-3
Prolonged release knocked down by the water curtain – on-site injuries	C3 – Serious – lost-time injuries on site	1.5E-4
Prolonged release drifts away from occupied areas – plant evacuated	C2 – Moderate – first aid / medical treatment on site	5.3E-5
Toxic cloud over occupied areas – people shelter, serious injuries	C4 – Major – serious injuries in occupied areas	1.2E-5
Toxic cloud – fatalities on and off site	C5 – Catastrophic – fatalities on and off site	1.3E-6

Category	Frequency /yr	Target /yr	Meets target?
C1 Minor – brief local exposure	5.1E-3	–	Yes
C2 Moderate – first aid / medical treatment on site	2.2E-3	1	Yes
C3 Serious – lost-time injuries on site	1.5E-4	0.1	Yes
C4 Major – serious injuries in occupied areas	1.2E-5	0.01	Yes
C5 Catastrophic – fatalities on and off site	1.3E-6	0.0001	Yes

The off-site toxic fatality frequency of 1.3E-6 /yr is compared with the 1E-4 /yr target; the water curtain (R-013) and the refuge (R-016) carry most of the mitigation.

8.2 ETA-AM-02 – Synthesis gas release at the synthesis loop

Initiating event = top event of bow tie BT-AM-02, frequency 9.6E-3 /yr. Hydrogen ignites more easily than natural gas: immediate ignition 0.3, delayed ignition 0.3. End states are reproduced by the bow-tie consequences JET, FF and VCE. HAZOP N03-04, N03-08. Illustrative probabilities.



ETA-AM-02 as drawn by Event Tree Analysis.

Heading	Success / failure	P(failure)	Source
H1 No immediate ignition	No immediate ignition / Immediate ignition	3.0E-1	High-pressure hydrogen ignites easily – illustrative; = BT-AM-02 'JET' condition
H2 Loop isolated and depressurised by ESD	Isolated / Not isolated	1.0E-1	= BT-AM-02 barrier 'Unit ESD and emergency depressurisation' (library ipl-sil1)
H3 No delayed ignition	No ignition / Delayed ignition	3.0E-1	= BT-AM-02 barrier 'Ignition source control'
H4 No vapour cloud explosion	Flash fire only / VCE	4.0E-1	Congested compressor house – illustrative

End state	Category	Frequency /yr
Release isolated – hydrogen disperses without ignition	C1 – Minor – gas released and dispersed	4.2E-3
Small flash fire – isolated inventory	C2 – Moderate – small flash fire	1.8E-3
Prolonged release disperses without ignition	C1 – Minor – gas released and dispersed	4.7E-4
Large flash fire – prolonged release	C4 – Major – prolonged jet fire or large flash fire	1.2E-4
Vapour cloud explosion – blast damage, multiple fatalities	C5 – Catastrophic – vapour cloud explosion	8.1E-5
Jet fire – isolated quickly, local damage	C3 – Serious – short jet fire	2.6E-3
Prolonged jet fire – escalation to the loop	C4 – Major – prolonged jet fire or large flash fire	2.9E-4

Category	Frequency /yr	Target /yr	Meets target?
C1 Minor – gas released and dispersed	4.7E-3	–	Yes
C2 Moderate – small flash fire	1.8E-3	1	Yes
C3 Serious – short jet fire	2.6E-3	0.1	Yes
C4 Major – prolonged jet fire or large flash fire	4.1E-4	0.01	Yes

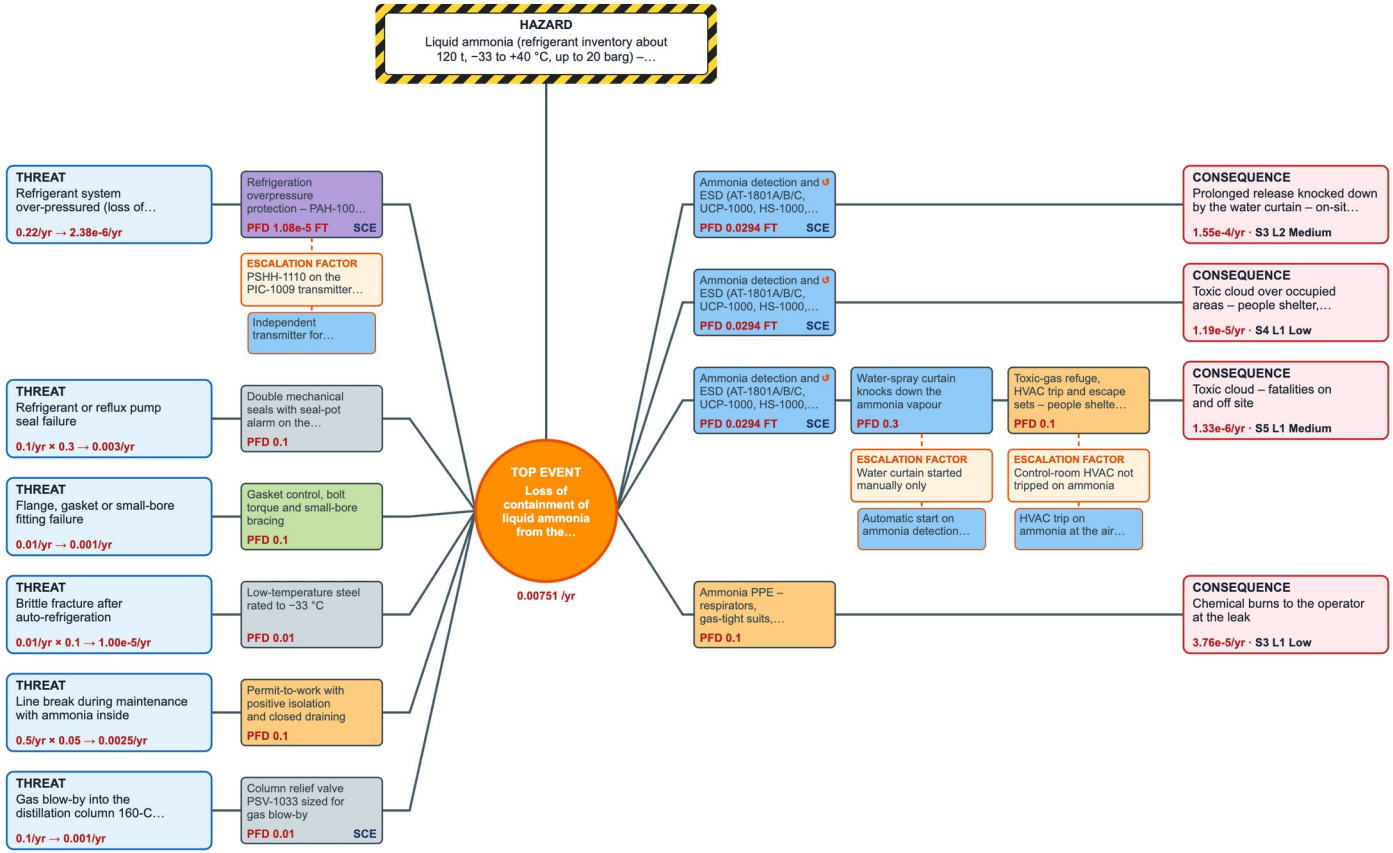
Category	Frequency /yr	Target /yr	Meets target?
C5 Catastrophic – vapour cloud explosion	8.1E-5	0.0001	Yes

The vapour cloud explosion frequency of 8.1E-5 /yr reflects the high ignition probability of hydrogen; automatic depressurisation (R-019) is the main lever.

9 Bow tie analysis

9.1 BT-AM-01 – Loss of containment of liquid ammonia from the refrigeration and recovery systems

Threat T1 frequency = FT-AM-01 demand gate G-006 with the overpressure barrier = FT-AM-01 gate G-002 (its contribution equals the FT-AM-01 top event). The ESD barrier = FT-AM-03 gate G-001. The top-event frequency is the initiating frequency of event tree ETA-AM-01, and consequences TOXON, TOXOCC and TOXOFF reproduce its end states. HAZOP N04, N05; FMECA S04, S05. Illustrative data.



BT-AM-01 as drawn by Bow Tie Analysis.

Hazard: Liquid ammonia (refrigerant inventory about 120 t, −33 to +40 °C, up to 20 barg) – toxic, IDLH 300 ppm. Top event: Loss of containment of liquid ammonia from the refrigeration or recovery system – frequency 7.5E-3 /yr.

Threat	Frequency /yr	Condition	Barriers	Contribution /yr
Refrigerant system over-pressured (loss of condensing)	2.2E-1	1	Refrigeration overpressure protection – PAH-1009 + operator, PSHH-1110 trip and PSV-1109	2.4E-6
Refrigerant or reflux pump seal failure	1.0E-1	0.3	Double mechanical seals with seal-pot alarm on the refrigerant pumps	3.0E-3
Flange, gasket or small-bore fitting failure	1.0E-2	1	Gasket control, bolt torque and small-bore bracing	1.0E-3
Brittle fracture after auto-refrigeration	1.0E-2	0.1	Low-temperature steel rated to −33 °C	1.0E-5
Line break during maintenance with ammonia inside	5.0E-1	0.05	Permit-to-work with positive isolation and closed draining	2.5E-3
Gas blow-by into the distillation column 160-C (let-down fails open)	1.0E-1	1	Column relief valve PSV-1033 sized for gas blow-by	1.0E-3

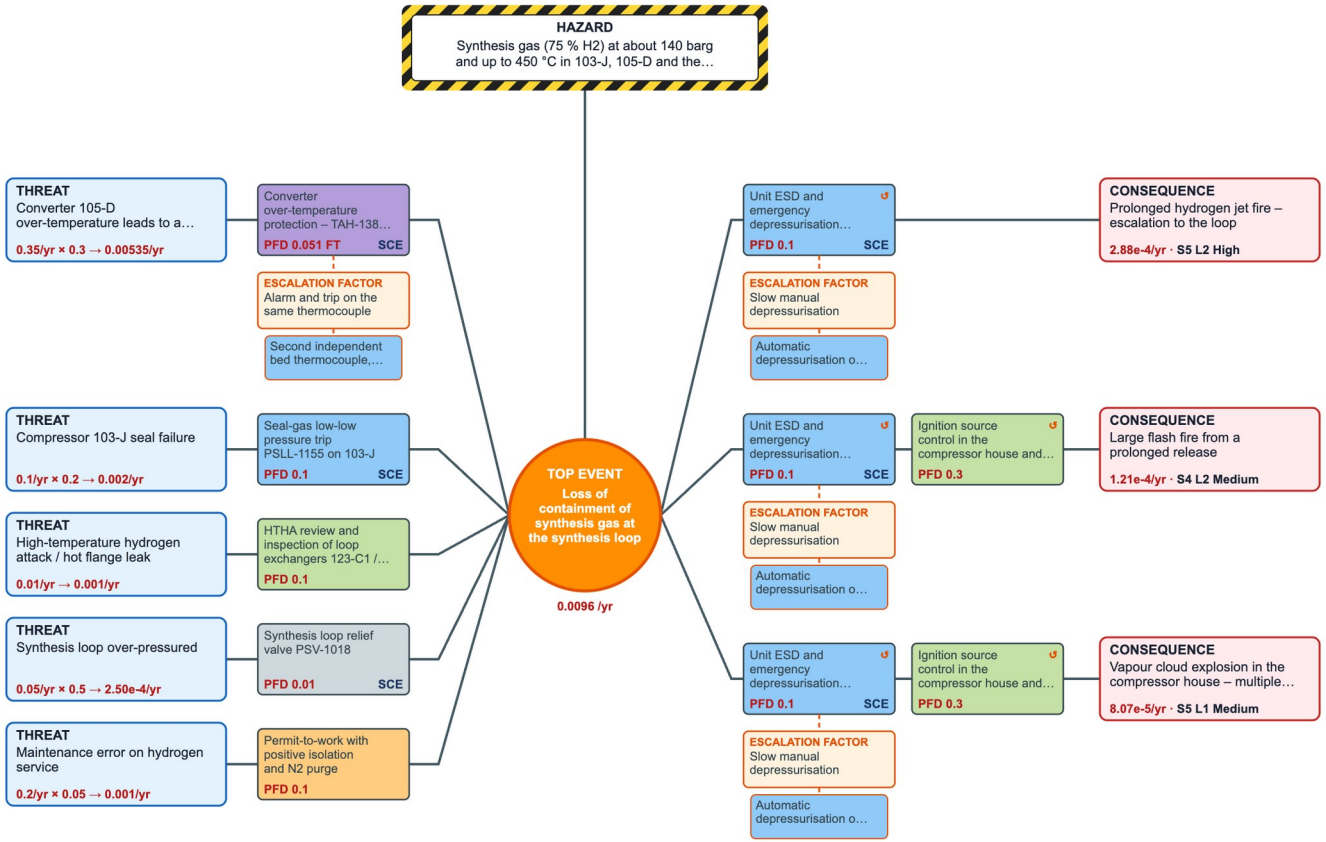
Barrier	Type	PFD	Performance standard	Verification	Critical
Refrigeration overpressure protection – PAH-1009 + operator, PSHH-1110 trip and PSV-1109	Active human-hardware	1.1E-5	Receiver never above 20 barg	Annual trip test; PSV-1109 bench test every 2 years	Yes
Double mechanical seals with seal-pot alarm on the refrigerant pumps	Passive hardware	1.0E-1	Primary seal failure contained	Monthly seal-pot check	
Gasket control, bolt torque and small-bore bracing	Continuous / procedural	1.0E-1	Joints made to procedure	Annual leak survey	
Low-temperature steel rated to −33 °C	Passive hardware	1.0E-2	No brittle fracture on auto-refrigeration	Material records (R-014)	
Permit-to-work with positive isolation and closed draining	Active human	1.0E-1	No line break with ammonia inside	PTW audits	
Column relief valve PSV-1033 sized for gas blow-by	Passive hardware	1.0E-2	Column never above 20 barg	Bench test	Yes
Ammonia detection and ESD (AT-1801A/B/C, UCP-1000, HS-1000, XV-1290 / XV-1801)	Active hardware	2.9E-2	Release isolated within 2 minutes	Quarterly detector bump tests; annual valve tests	Yes
Water-spray curtain knocks down the ammonia vapour	Active hardware	3.0E-1	Curtain running within 2 minutes	Annual flow test (R-013)	
Toxic-gas refuge, HVAC trip and escape sets – people shelter in time	Active human	1.0E-1	Everyone in refuge within 5 minutes	Annual drill (R-016)	
Ammonia PPE – respirators, gas-tight suits, eyewash and showers	Active human	1.0E-1	No unprotected exposure	Supervision	

Consequence	Condition	Barriers	Residual /yr	Risk rank
Prolonged release knocked down by the water curtain – on-site injuries	0.7	Ammonia detection and ESD (AT-1801A/B/C, UCP-1000, HS-1000, XV-1290 / XV-1801)	1.5E-4	Medium
Toxic cloud over occupied areas – people shelter, serious injuries	0.054	Ammonia detection and ESD (AT-1801A/B/C, UCP-1000, HS-1000, XV-1290 / XV-1801)	1.2E-5	Low
Toxic cloud – fatalities on and off site	0.2	Ammonia detection and ESD (AT-1801A/B/C, UCP-1000, HS-1000, XV-1290 / XV-1801) Water-spray curtain knocks down the ammonia vapour Toxic-gas refuge, HVAC trip and escape sets – people shelter in time	1.3E-6	Medium
Chemical burns to the operator at the leak	0.05	Ammonia PPE – respirators, gas-tight suits, eyewash and showers	3.8E-5	Low

Seal and gasket failures (T2, T3) dominate the ammonia top event; refrigeration overpressure (T1) is small once the overpressure protection is credited.

9.2 BT-AM-02 – Loss of containment of hydrogen-rich synthesis gas at the synthesis loop

Threat T1 = converter over-temperature (FT-AM-02 demand gate G-006) with the protection barrier = FT-AM-02 gate G-002 and 0.3 = probability that the excursion causes a leak. The top-event frequency is the initiating frequency of event tree ETA-AM-02; consequences JET, FF and VCE reproduce its end states. HAZOP N03; FMECA S03. Illustrative data.



BT-AM-02 as drawn by Bow Tie Analysis.

Hazard: Synthesis gas (75 % H2) at about 140 barg and up to 450 °C in 103-J, 105-D and the loop exchangers. Top event: Loss of containment of synthesis gas at the synthesis loop – frequency 9.6E-3 /yr.

Threat	Frequency /yr	Condition	Barriers	Contribution /yr
Converter 105-D over-temperature leads to a flange or exchanger leak	3.5E-1	0.3	Converter over-temperature protection – TAH-1383 + operator and TSHH-1396 trip	5.4E-3
Compressor 103-J seal failure	1.0E-1	0.2	Seal-gas low-low pressure trip PSLL-1155 on 103-J	2.0E-3
High-temperature hydrogen attack / hot flange leak	1.0E-2	1	HTHA review and inspection of loop exchangers 123-C1 / 124-C1	1.0E-3
Synthesis loop over-pressured	5.0E-2	0.5	Synthesis loop relief valve PSV-1018	2.5E-4
Maintenance error on hydrogen service	2.0E-1	0.05	Permit-to-work with positive isolation and N2 purge	1.0E-3

Barrier	Type	PFD	Performance standard	Verification	Critical
Converter over-temperature protection – TAH-1383 + operator and TSHH-1396 trip	Active human-hardware	5.1E-2	Catalyst never above 520 °C	Annual trip test (R-008)	Yes

Barrier	Type	PFD	Performance standard	Verification	Critical
Seal-gas low-low pressure trip PSLL-1155 on 103-J	Active hardware	1.0E-1	Compressor trips before seal failure	Annual trip test	Yes
HTHA review and inspection of loop exchangers 123-C1 / 124-C1	Continuous / procedural	1.0E-1	No HTHA damage in service	Turnaround inspection (R-010)	
Synthesis loop relief valve PSV-1018	Passive hardware	1.0E-2	Loop never above 160 barg	Bench test	Yes
Permit-to-work with positive isolation and N2 purge	Active human	1.0E-1	No break-in on hydrogen service without isolation	PTW audits	
Unit ESD and emergency depressurisation (UCP-1000, HV-1019)	Active hardware	1.0E-1	Loop isolated and depressurised to flare	Annual ESD test	Yes
Ignition source control in the compressor house and loop area	Continuous / procedural	3.0E-1	No ignition source in zoned areas	Ex inspection	

Consequence	Condition	Barriers	Residual /yr	Risk rank
Prolonged hydrogen jet fire – escalation to the loop	0.3	Unit ESD and emergency depressurisation (UCP-1000, HV-1019)	2.9E-4	High
Large flash fire from a prolonged release	0.42	Unit ESD and emergency depressurisation (UCP-1000, HV-1019) Ignition source control in the compressor house and loop area	1.2E-4	Medium
Vapour cloud explosion in the compressor house – multiple fatalities	0.28	Unit ESD and emergency depressurisation (UCP-1000, HV-1019) Ignition source control in the compressor house and loop area	8.1E-5	Medium

Compressor seal failure (T2) and hot-flange / HTHA leaks (T3) dominate; the converter over-temperature threat (T1) depends on R-008.

10 How the tools link – cross-reference

The cross-link tests (section 11) prove each of these links on the files themselves.

From	To	What is shared
HAZID HZD-07, HZD-11, HZD-12	HAZOP N03-03, N04-01, N04-07	Hazards sent to HAZOP nodes
HAZOP N04-01	FT-AM-01	Refrigeration overpressure; likelihood from the FT top event
HAZOP N03-03	FT-AM-02	Converter over-temperature; likelihood from the FT top event
FT-AM-01 G-002	BT-AM-01 barrier REFOP	Overpressure protection PFDavg (live link)
FT-AM-01 G-006	BT-AM-01 threat T1	Overpressure demand frequency
FT-AM-03 G-001	ETA-AM-01 H1 and BT-AM-01 barrier ESD	ESD PFDavg (live links)
BT-AM-01 / BT-AM-02 top events	ETA-AM-01 / ETA-AM-02 initiating events	Release frequencies
ETA-AM-01 end states	BT-AM-01 consequences TOXON, TOXOCC, TOXOFF	Same frequencies
ETA-AM-02 end states	BT-AM-02 consequences JET, FF, VCE	Same frequencies
FT-AM-02 G-002 / G-006	BT-AM-02 barrier CONVPROT / threat T1	Converter protection and demand
FMECA S01–S05	HAZOP N01–N05, FT events	Same tags and failure modes

10.1 Key results

Value	Result
Receiver over-pressured beyond design /yr (FT-AM-01)	2.4E-6
Overpressure protection PFDavg	1.1E-5
Overpressure demand /yr	2.2E-1
Unprotected converter over-temperature /yr (FT-AM-02)	1.8E-2
Converter protection PFDavg	5.1E-2
Over-temperature demand /yr	3.5E-1
Ammonia detection & ESD PFDavg (FT-AM-03)	2.9E-2
Ammonia detection & ESD RRF	3.4E1
Ammonia loss of containment /yr (BT-AM-01)	7.5E-3
BT1_TOXON	1.5E-4
BT1_TOXOCC	1.2E-5
BT1_TOXOFF	1.3E-6
BT1_BURN	3.8E-5
Synthesis gas loss of containment /yr (BT-AM-02)	9.6E-3
BT2_JET	2.9E-4
BT2_FF	1.2E-4
BT2_VCE	8.1E-5
Toxic fatalities on and off site /yr (ETA-AM-01)	1.3E-6
Serious injuries in occupied areas /yr	1.2E-5
Vapour cloud explosion /yr (ETA-AM-02)	8.1E-5
Prolonged jet fire or large flash fire /yr	4.1E-4

11 Verification of the example (cross-link tests)

Every study file was re-opened in its toolkit app and recalculated by the app's own engine; the workbooks were recalculated in LibreOffice. The automated test (test_example.py) ran 556 checks: 556 passed, 0 failed.

Check group	Passed	Failed
A. Apps	34	0
B. Numbers	70	0
C. References	138	0
D. Tags	39	0
E. Workbooks	83	0
G. HAZID	87	0
W. What-If	35	0
L. LOPA	14	0
F. Report	56	0

12 Recommendations register

No.	Recommendation	Responsible	Priority	HAZOP refs
R-001	Confirm that the low steam-to-carbon trip FFSLL-1201 and the low steam trip	Instrument / SIS	High	N01-01, N01-

No.	Recommendation	Responsible	Priority	HAZOP refs
	FSLL-1201 are independent of the ratio controller FFI-1001 (separate transmitters) and SIL-assessed.	engineer		02, N01-09
R-002	Install tube-wall temperature monitoring (fixed IR scanners) on the primary reformer 101-B and add tube-wall limits to the operating window.	Process engineer	Medium	N01-04
R-003	Review the process-air trip FFSHH-1203 response time against the secondary reformer 103-D over-temperature scenario.	Process engineer	Medium	N01-06
R-004	Verify the reformer BMS flame-failure function against IEC 61511 (scanners per burner row, proof-test interval) – the arch burners 101-BBA run at low turndown during start-up.	Instrument / SIS engineer	High	N02-01, N02-02, N02-06
R-005	Route the purge gas from N05 to the reformer fuel only through the knock-out drum with an ammonia analyser and automatic diversion to the flare on high ammonia.	Process engineer	Medium	N02-04, N05-01
R-006	Provide differential-pressure indication across the suction strainers of 103-J and 105-J (source recommendation 89).	Process engineer	Medium	N03-02, N04-04
R-007	Consult the compressor vendor on the response to suction-strainer choking leading to low suction pressure (source recommendation 104).	Rotating equipment engineer	Medium	N03-02, N04-04
R-008	Upgrade the converter over-temperature protection: fault tree FT-AM-02 gives an uncontrolled over-temperature frequency of 1.8E-2 /yr with protection PFDavg 5.1E-2; add a second independent bed thermocouple to TSHH-1396 (1oo2) and a six-monthly trip test.	Instrument / SIS engineer	High	N03-03
R-009	Confirm the loop relief PSV-1018 is sized for the start-up heater 102-B fire case and for blocked outlet with 103-J at maximum speed.	Process engineer	High	N03-04
R-010	Carry out a high-temperature hydrogen attack (HTHA) review of the loop exchangers 123-C1 / 124-C1 against the current Nelson curves.	Materials engineer	High	N03-06
R-011	Refrigeration overpressure: fault tree FT-AM-01 gives 2.4E-6 /yr with protection PFDavg 1.1E-5, but almost all of the credit is PSV-1109 – every demand ends as an ammonia relief to flare because the alarm and the trip share one transmitter; make PSHH-1110 independent of PIC-1009 and add an automatic start of the standby cooling-water pump.	Instrument / SIS engineer	High	N04-01
R-012	Verify the ammonia detection and ESD (FT-AM-03 PFDavg 2.9E-2): detectors at the receiver 149-D, the refrigerant pumps and the compressor house, 1oo3 with automatic isolation of XV-1290.	Instrument / SIS engineer	High	N04-07, N05-06
R-013	Extend the water-spray curtain to the refrigerant pump bay and test it annually; event tree ETA-AM-01 gives 1.3E-6 /yr for off-site toxic impact.	Fire officer	High	N04-07
R-014	Confirm all refrigeration piping and the flash drums are rated for −33 °C including depressurisation (auto-refrigeration of ammonia).	Materials engineer	Medium	N04-05
R-015	Add a low-low cooling-water flow trip at condenser 127-C that unloads 105-J before PSV-1109 lifts.	Process engineer	Medium	N04-01, N04-06
R-016	Provide a toxic-gas refuge procedure and annual drill for the ammonia area; confirm HVAC trip on ammonia at the control room.	HSE / process safety	High	N04-07
R-017	Add high-level protection LAH-1049 / LSHH on column 160-C to prevent liquid ammonia carry-over into the purge gas to fuel.	Process engineer	Medium	N05-03
R-018	Confirm that the purge let-down PV-1033A fail-open (gas blow-by) case is covered by PSV-1033 (160 barg into a 20 barg column).	Process engineer	High	N05-02, N05-04
R-019	Syngas release outcomes: event tree ETA-AM-02 gives 8.1E-5 /yr for a vapour cloud explosion at the synthesis loop; review congestion and the remote depressurisation HV-1019 response time.	Process engineer	High	N03-08
R-020	Add a start-up permissive: 102-B start-up heater may only fire with proven loop circulation (103-J running) – overheating of the converter at start-up.	Instrument / SIS engineer	Medium	N03-01, N03-09

13 Conclusions

- The source HAZOP was thorough on causes and safeguards but gave no risk ranking; ranking and quantification show where the protection is thin.
- In both the refrigeration system and the converter, the alarm and the trip share one sensor. At the converter this leaves an unprotected over-temperature frequency of 1.8E-2 /yr (R-008 is the priority action); in the refrigeration system the relief valve keeps the overpressure frequency at 2.4E-6 /yr, but every demand ends as a toxic relief (R-011).
- A large ammonia release gives an off-site fatality frequency of 1.3E-6 /yr, provided the detection and ESD (PFDavg 2.9E-2), the water curtain and the refuge perform as assumed – R-012, R-013 and R-016.
- At the synthesis loop, hydrogen's high ignition probability drives the explosion frequency (8.1E-5 /yr); automatic depressurisation (R-019) and HTHA inspection (R-010) reduce it.

14 Limitations

- All design data and failure data are illustrative. Library values marked 'typical' must be verified against the edition used; site and vendor data take precedence.
- Consequence severities are judgement; no toxic dispersion, fire or explosion modelling was done.

- Only five of the source's nodes are taken forward; the risk ranking and quantified studies are added for this example.